

Making Transformer interpretable by design: A self-representation architecture for multi-scenario fine-grained land cover mapping[☆]

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ABSTRACT

Fine-grained land cover mapping leveraging deep learning and hyperspectral remote sensing imagery (HRSI) plays an important role in a wide range of application scenarios. Currently, Transformer has shown strong potential in this task due to its global modeling capability. However, despite its promising performance, the limited interpretability remains a serious concern for model reliability. To bridge this gap, we propose an inherently interpretable self-representation Transformer (SRTF) that integrates the mathematical reasoning of representation models into the Transformer architecture design. More specifically, the proposed SRTF replaces the two most opaque components in Transformer, the self-attention mechanism and the multilayer perceptron, with theoretically grounded self-representation mechanism (SRM) and class subspace learning (CSSL), respectively. First, SRM is inspired by the optimization principle of sparse representation, constructing long-range dependencies using the input itself as the dictionary. Meanwhile, to enable parallel interactions among local features, SRM is further extended to a grouped version. Second, the CSSL module is developed based on the subspace decomposition process to extract discriminative subspace components, with a fidelity loss imposed to guarantee valid decomposition. Moreover, to enhance the transparency of the decision-making process, each subspace component in the final CSPL module is explicitly associated with a specific class, enabling each decision to be traced back to its corresponding feature subspace. Experiments were conducted across multiple land cover scenarios in China, each with at least 20 fine-grained classes. Results and analyses demonstrate that the proposed SRTF, while providing enhanced interpretability, also delivers strong practical performance and consistently outperforms related methods in both accuracy and generalization capability.

1. Introduction

Land cover mapping is a fundamental task in Earth science research, with its primary objective being to identify and distinguish various types of surface cover, providing critical insights into the evolution of land surface patterns and supporting environmental monitoring and resource management (Verburg et al., 2011; Song et al., 2018; Winkler et al., 2021). In recent years, with continuous advancements in Earth observation and remote sensing imaging technologies, land cover mapping has received reliable data support, while the precision of surface feature interpretation has been significantly enhanced (Friedl et al., 2002; Gong et al., 2019; Brown et al., 2022; Wang et al., 2023). Among the available remote sensing data sources, hyperspectral remote sensing imagery (HRSI), which provides continuous spectral measurements capable of capturing subtle differences among surface

objects, has become a critical resource for high-precision and fine-grained land cover mapping (Vali et al., 2020; Huang et al., 2023a; Moharram and Sundaram, 2023; Lou et al., 2025).

As illustrated in Fig. 1, to meet the fine-grained mapping requirements of different application scenarios such as agriculture, forestry, wetland, and built-up area, HRSI acquired from satellite or unmanned aerial vehicle (UAV) platforms is widely employed. By exploiting its continuous spectral information and rich spatial attributes, different land cover types can be effectively distinguished, enabling high-accuracy surface classification (Patro et al., 2021). Depending on the observation platform, satellite-borne HRSI offers extensive spatial coverage suitable for large-scale mapping, (Su et al., 2021; Huang et al., 2023b), whereas UAV-borne HRSI provides limited coverage

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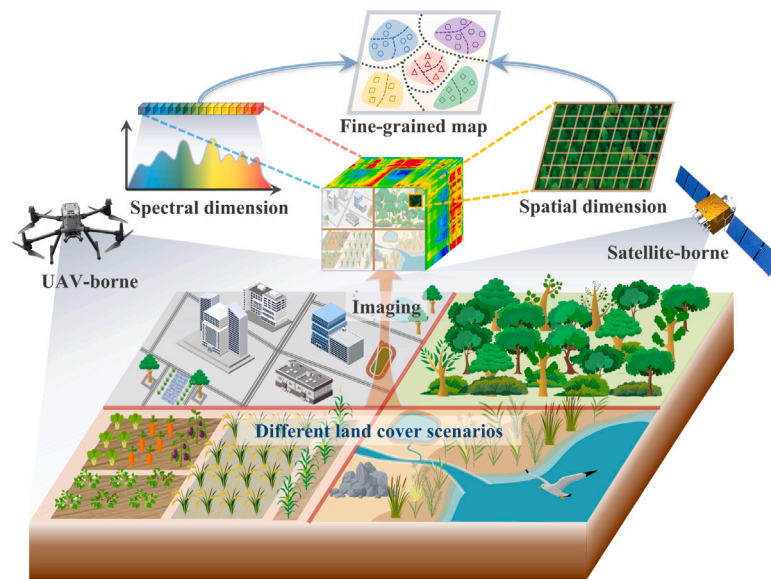


Fig. 1. Schematic illustration of fine-grained land cover mapping using satellite-borne or UAV-borne hyperspectral remote sensing imagery, leveraging spectral and spatial information for accurate classification across diverse land cover scenarios.

but delivers much higher spatial resolution that supports refined characterization of local surface features (Zhong et al., 2020; Hu et al., 2022). With the reliable data foundation provided by HRSI, continuous advancements in classification methods are essential for improving surface identification accuracy and advancing land cover mapping research toward greater automation and intelligence. From a comprehensive perspective of methodology and development stages, existing HRSI mapping methods can be broadly divided into three mainstreams, including classical machine learning, representation models, and deep learning.

Early HRSI-based land cover mapping studies primarily relied on classical machine learning methods (Du et al., 2012), such as the widely utilized support vector machine (SVM) (Xia et al., 2016) and random forest (RF) (Belgiu and Drăguț, 2016), favored for their intuitive principles and ease of implementation. However, by focusing only on pixel-wise spectral information and disregarding spatial context, these methods often yield maps with poor spatial continuity and limited accuracy (Bioucas-Dias et al., 2013). Moreover, when applied to HRSI with hundreds of spectral bands, their performance is constrained by the curse of dimensionality (Huang et al., 2019). To mitigate these limitations, manually engineered spatial features and dimensionality reduction techniques are commonly employed (Kumar et al., 2020; Patro et al., 2021). Nevertheless, these additional computational steps depend heavily on expert knowledge, and their effectiveness is often constrained by scenario-specific characteristics and varying imaging conditions.

Subsequently, with the rapid advancement of representation models in pattern recognition and computer vision, these models exert a profound impact on the interpretation of remote sensing imagery, particularly for HRSI-based land cover mapping (Peng et al., 2022). Common representation models include sparse representation (SR) (Zhang et al., 2021), collaborative representation (CR) (Su et al., 2023), and low-rank representation (LRR) (Han et al., 2025). Compared with classical machine learning methods, representation models offer several distinct advantages. On the one hand, through the way of joint representation, these models can effectively integrate spectral and spatial information (Peng and Du, 2017), enhancing the spatial continuity and accuracy of the mapping results. On the other hand, representation models leverage sparsity or low-rank constraints to capture the latent low-dimensional manifold structure of high-dimensional data (Li et al., 2016), thereby effectively mitigating the impact of the inherent curse of dimensionality

in HRSI processing. Moreover, as model-driven methods grounded in rigorous optimization principles, representation models provide strong theoretical interpretability, while demonstrating strong applicability and generalization capability in land cover mapping tasks (Xue et al., 2017; Zhang et al., 2017; Su et al., 2021). Nevertheless, several fundamental limitations continue to hinder the broader applicability of representation models: (1) The optimization process of these models is computationally intensive, often requiring hundreds of iterations to achieve convergence; (2) Performance critically depends on manually tuned regularization parameters, rendering the tuning process complex and highly reliant on expert experience; (3) The linear representation process based on dictionary atoms restricts their ability to capture nonlinear relationships and higher-order semantic patterns in HRSI data.

In recent years, data-driven deep learning methods have shown remarkable potential in HRSI processing (Vali et al., 2020; Kumar et al., 2024; Song et al., 2025). Leveraging end-to-end automatic feature learning and strong nonlinear modeling capabilities, deep learning-based methods can automatically extract discriminative spatial-spectral representations and adaptively optimize model parameters through the backpropagation process (Dong et al., 2022; Wang et al., 2025a; He et al., 2026). Among existing network architectures, Transformer has emerged as one of the most representative and influential, distinguished by its self-attention mechanism (SAM) that effectively captures long-range dependencies and enables global modeling of spectral and spatial information (Mei et al., 2022; X. Li et al., 2024). To better adapt to the characteristics of HRSI data, numerous improved Transformer models have been proposed. For instance, Hong et al. (2022) introduced SpectralFormer, a Transformer-based backbone that models local spectral sequences across neighboring bands and employs cross-layer skip connections to effectively capture the sequential nature of spectral signatures. Subsequently, Yang et al. (2023) adopted a hierarchical region sampling strategy and integrated center and surrounding Transformers to achieve multi-scale and detail-aware feature learning, effectively addressing challenges related to central pixel contribution ambiguity and limited spatial interactions. Later, for efficient feature extraction and lightweight network design, Zhao et al. (2024) incorporated separable convolution and SAM modules, and replaced the multilayer perceptron (MLP) with a simplified pointwise layer enhanced by skip connections, significantly improving computational efficiency. Furthermore, recent research on Transformer-based methods has further evolved toward

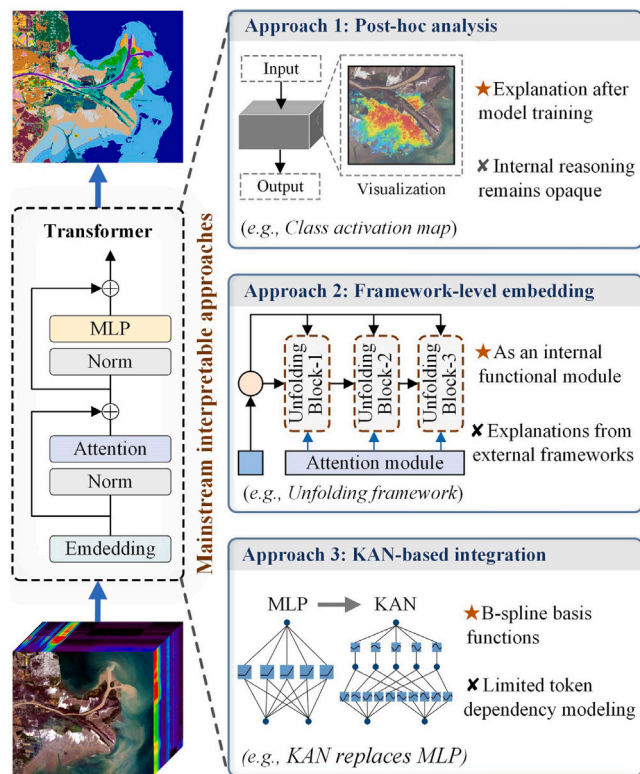


Fig. 2. Existing approaches to improve the interpretability of Transformer for land cover mapping and their limitations.

hybrid architectures (Cheng et al., 2024; Fu et al., 2025), integration of self-supervised learning (He et al., 2023; Z. Li et al., 2024), frequency-spatial domain transformations (Wang et al., 2024; Shi et al., 2025), and multimodal fusion (Liu et al., 2024; Wittstruck et al., 2025).

Although Transformer-based methods have significantly advanced land cover analysis, their reasoning process remains poorly interpretable. Notably, while the key functional component SAM possesses a well-defined mathematical formulation, it was not theoretically derived beforehand and instead represents a formal summary of effective empirical designs (Yu et al., 2024). Moreover, most current studies focus on enhancing performance within this opaque architecture through empirical improvements without substantially enhancing its interpretability. However, for management agencies and policymakers, understanding how classification decisions are made is essential for establishing confidence in model reliability and supporting informed land use planning and policy development. To address this gap, as shown in Fig. 2, a limited but steadily growing body of research has sought to enhance the transparency of Transformer through various approaches: (1) Applying post-hoc analysis techniques to interpret trained models. For example, Hu et al. (2023) visualizes class activation maps to identify image regions that influence model decisions and ranks key factors by importance, leading to an interpretable Transformer-based framework. Similarly, Kuang et al. (2025) demonstrates interpretability by analyzing the impact of input features on predictions and visualizing hidden-layer outputs; (2) Transformer modules are embedded within inherently interpretable frameworks as auxiliary functional components. In this strategy, the Transformer is integrated into a model with a transparent computational mechanism to enhance specific functionalities, while the interpretability of the overall framework is preserved. For example, Wang et al. (2025) incorporates a region-aware Transformer block into a deep unfolding framework, enabling global contextual modeling while maintaining the interpretability of the unfolding process; (3) Integrating Transformer with the Kolmogorov–Arnold network (KAN).

Relevant studies replace the conventional MLP module with KAN, whose learnable spline-based activation functions provide explicit representations of feature transformations (Xia et al., 2025; Zhang et al., 2025).

Despite the promising progress of the aforementioned approaches, several fundamental limitations remain: (1) Post-hoc analyses only provide passive explanations for the model after training, without altering the inherently opaque internal reasoning; (2) Framework-level methods derive interpretability primarily from the external model architecture, while the embedded Transformer modules themselves remain opaque and lack explicit interpretability; (3) KAN-based designs primarily target the transformation module and offer limited insight into the dynamic modeling of token dependencies. Unlike existing approaches, we endow Transformer with inherent interpretability in an interpretable-by-design manner. Specifically, we rethink the theoretical foundations and internal construction of key modules within this widely validated architecture. By grounding the reasoning process in mathematical operations from representation models, the proposed method overcomes the black-box limitation while preserving high performance in land cover mapping. The main contributions of this study are summarized as:

- A novel self-representation Transformer (SRTF) network is proposed for HRSI-based land cover mapping. By embedding the mathematical foundations of representation models into the Transformer architecture design, SRTF enhances interpretability while maintaining strong practical performance.
- We develop two theoretically grounded modules, the self-representation mechanism (SRM) inspired by the optimization principle of sparse representation, and the class subspace learning (CSSL) derived from the subspace decomposition process. These modules replace the block-box designed SAM and MLP, respectively. Meanwhile, a group self-representation mechanism (GSRM) is developed to achieve parallel interactions among local features.
- To further enhance the transparency of the decision-making process, we explicitly associate each subspace component in the final CSSL module with a corresponding land cover class, enabling each classification decision to be directly traced back to its relevant feature subspace.

Experiments were conducted across four different study areas in China to evaluate the proposed method, each study area representing a distinct land cover scenario, with each scenario containing at least 20 fine-grained classes. Across all study areas, compared with methods of the same network architecture, our method achieves higher accuracy and demonstrates superior generalization performance.

2. Materials

2.1. Study areas

The locations of the study areas and their corresponding hyperspectral datasets are shown in Fig. 3, covering four representative land cover scenarios. Specifically, the HongHu dataset was acquired in Honghu City, Hubei Province, China, and encompasses a variety of crop types, representing a typical agricultural landscape. The Yellow River Delta dataset, collected from the Yellow River Delta National Nature Reserve in Dongying City, Shandong Province, corresponds to a typical wetland environment. The SZUTreeHSI dataset was collected in Shenzhen, Guangdong Province, focusing on the region covered with multiple tree species. Finally, the Hohai University dataset, captured in Changzhou, Jiangsu Province, characterizes a built-up area dominated by artificial structures such as buildings and roads.

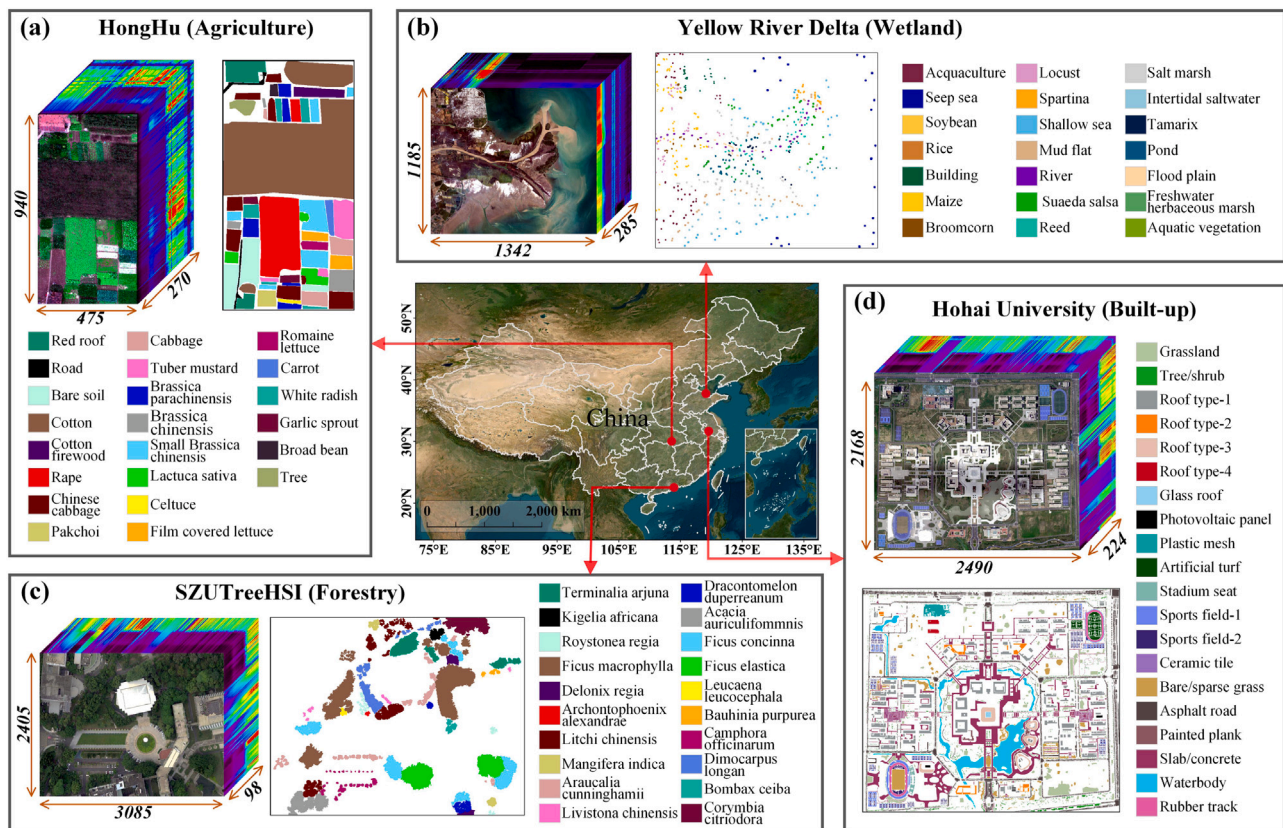


Fig. 3. Locations of the study areas across different land cover scenarios and their corresponding hyperspectral datasets. (a) HongHu (Agriculture), (b) Yellow River Delta (Wetland), (c) SZUTreeHSI (Forestry), (d) Hohai University (Built-up).

Table 1

Number of labeled samples for different classes in each hyperspectral dataset.

No.	HongHu		Yellow River Delta		SZUTreeHSI		Hohai University	
	Class	Samples	Class	Samples	Class	Samples	Class	Samples
1	Red roof	59,807	Acquaculture	393	Terminalia arjuna	108,973	Grassland	126,689
2	Road	14,041	Seep sea	796	Kigelia africana	20,046	Tree/shrub	62,654
3	Bare soil	3512	Soybean	110	Roystonea regia	22,322	Roof type-1	212,575
4	Cotton	21,821	Rice	190	Ficus macrophylla	412,152	Roof type-2	24,530
5	Cotton firewood	163,285	Building	83	Delonix regia	23,047	Roof type-3	38,185
6	Rape	6218	Maize	96	Archontophoenix alexandrae	2696	Roof type-4	7713
7	Chinese cabbage	44,557	Broomcorn	95	Litchi chinensis	139,666	Glass roof	2588
8	Pakchoi	24,103	Locust	211	MMangifera indica	22,755	Plastic mesh	3251
9	Cabbage	4054	Spartina	200	Araucalia cunninghamii	88,273	Photovoltaic panel	21,711
10	Tuber mustard	10,819	Shallow sea	936	Livistona chinensis	13,442	Artificial turf	29,741
11	Brassica parachinensis	12,394	Mud flat	553	Dracontomelon duperreanum	31,217	Stadium seat	82,49
12	Brassica chinensis	11,015	River	469	Acacia auriculiformis	86,333	Sports field-1	55,029
13	Small Brassica chinensis	8954	Suaeda salsa	361	Ficus concinna	194,931	Sports field-2	33,499
14	Lactuca sativa	22,507	Reed	240	Ficus elastica	165,964	Ceramic tile	8617
15	Celtuce	7356	Salt marsh	595	Leucaena leucocephala	5307	Bare/sparse grass	92,439
16	Film covered lettuce	1002	Intertidal saltwater	454	Bauhinia purpurea	8559	Asphalt road	295,857
17	Romaine lettuce	7262	Tamarix	133	Camphora officinarum	37,026	Painted plank	12,855
18	Carrot	3010	Pond	377	Dimocarpus longan	80,492	Slab/concrete	392,349
19	White radish	3217	Flood plain	68	Bombax ceiba	15,120	Waterbody	152,750
20	Garlic sprout	8712	Freshwater herbaceous marsh	72	Corymbia citriodora	86,534	Rubber track	28,062
21	Broad bean	3486	Aquatic vegetation	39				
22	Tree	1328						

2.2. Dataset details

2.2.1. Agriculture scenario dataset

The HongHu hyperspectral imagery (Zhong et al., 2020) was acquired using a Headwall Nano-Hyperspec imaging sensor (with a 17 mm focal length), mounted on a DJI Matrice 600 Pro UAV platform flying at an altitude of 100 m, on November 20, 2017. The imagery has a spatial resolution of 0.043 m with 940×475 pixels, covering

270 spectral bands within the 400–1000 nm wavelength range. As summarized in Table 1, the ground-truth map includes 22 land cover classes, providing a diverse basis for fine-scale classification.

2.2.2. Wetland scenario dataset

The Yellow River Delta hyperspectral imagery (Su et al., 2021) was acquired on January 7, 2019 by the Advanced Hyperspectral Imager

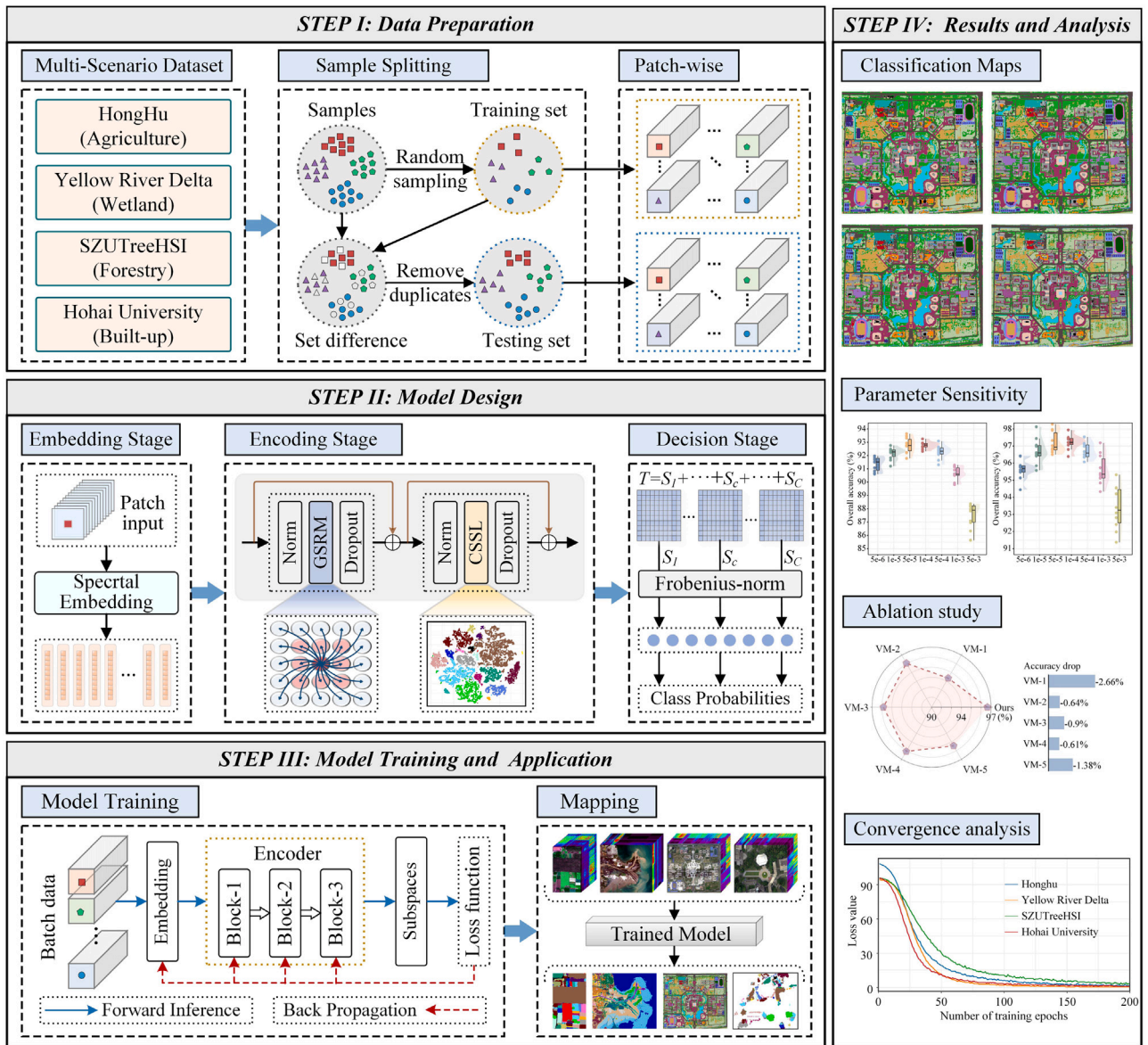


Fig. 4. Overall workflow of the study, comprising data preparation (Step I), model design (Step II), model training and application (Step III), and results analysis (Step IV).

(AHSI) onboard the Gaofen-5 satellite. The imagery has a spatial resolution of 30 m and a size of 1185×1342 pixels, covering 330 spectral bands spanning the visible to shortwave infrared (400–2500 nm) range. After removing 45 noise-contaminated bands, a total of 285 bands were retained for land cover mapping. The corresponding ground-truth map contains 21 land cover classes, with the number of labeled samples for each class listed in Table 1.

2.2.3. Forestry scenario dataset

The SZUTreeHSI hyperspectral imagery (N. Li et al., 2024) was acquired using a UAV-mounted SPECIM FX10 imaging system. The imagery has a spatial resolution of 0.1 m and a spectral resolution of 5.5 nm, with an image size of 2405×3085 pixels. After excluding noise-contaminated bands, a total of 98 spectral bands were retained, spanning the wavelength range from 399 nm to 924 nm. The corresponding ground-truth map of the imagery includes 20 different tree species classes, and Table 1 provides detailed information on the samples for each class.

2.2.4. Built-up area scenario dataset

The Hohai University hyperspectral imagery was acquired in July 23, 2024, using the GaiaSky-mini airborne hyperspectral imaging system, which was mounted on a DJI Matrice M300 RTK UAV platform. The imagery was originally captured at a spatial resolution of 0.26 m and was subsequently resampled to 0.5 m for processing consistency. It has a spatial size of 2168×2490 pixels and contains 224 spectral bands, covering the 400–1000 nm wavelength range. The corresponding ground-truth map includes 20 distinct land cover classes, and the number of labeled samples for each class is also provided in Table 1.

3. Methods

The overall workflow of this study is illustrated in Fig. 4 and consists of four main steps. Step I involves data preparation, including the collection of multi-scenario hyperspectral datasets, the partitioning of training and test sets, and the construction of patch-wise samples. Step II presents the model design process, with the proposed SRTF network described in detail in this section. Step III describes model training and the application of the trained model to land cover mapping. Step IV

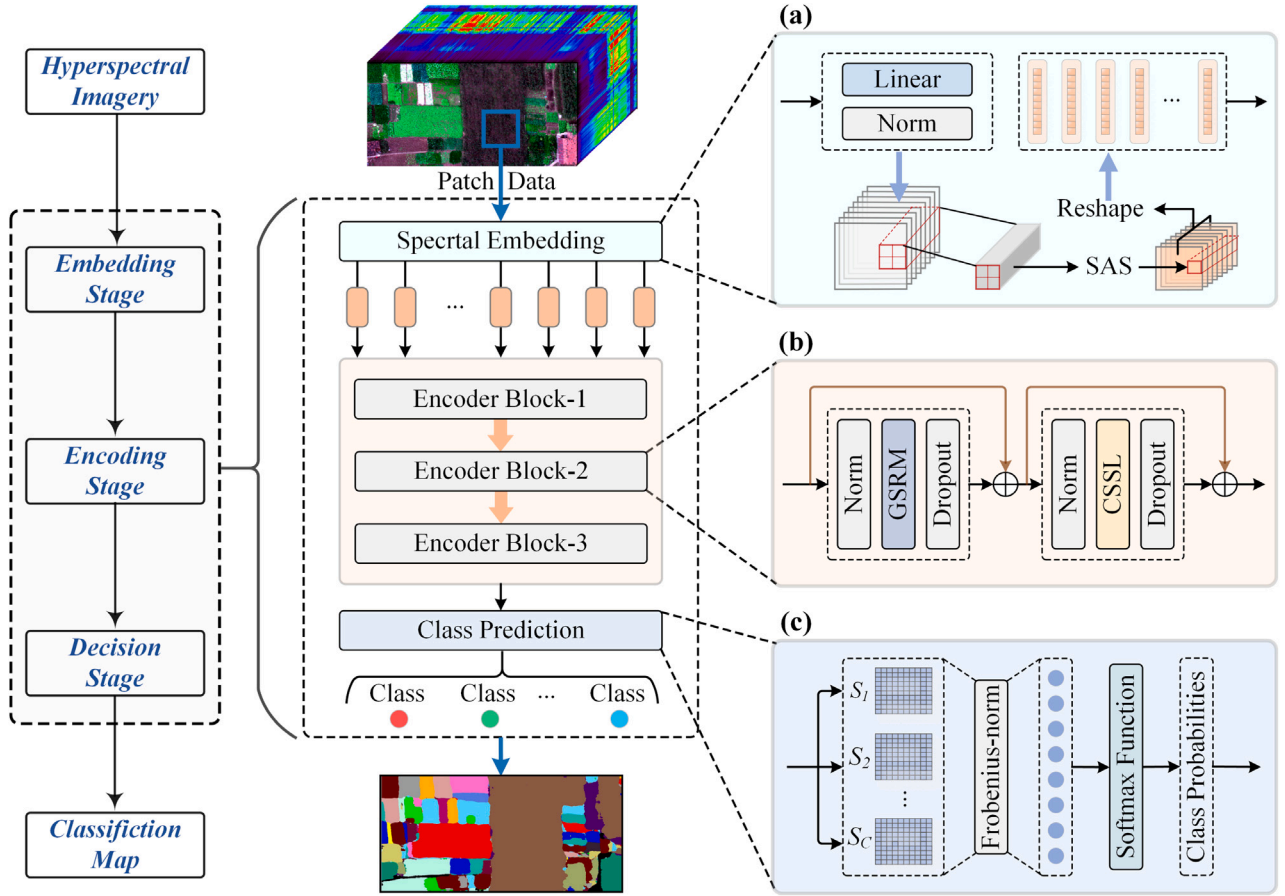


Fig. 5. Flowchart of the proposed SRTF for HRSI classification. Subfigures (a), (b), and (c) describe spectral embedding, encoder block, and class prediction in detail, respectively.

evaluates the mapping results and conducts experimental analyses to verify the effectiveness of the proposed method.

As illustrated in Fig. 5, SRTF performs classification through three main stages, namely embedding stage, encoding stage, and decision stage. These stages are implemented by the spectral embedding part, the encoder part, and the class prediction part, respectively. In the embedding stage, the input 3D patch data are projected into a specific latent space and then reshaped into a token sequence, which serves as the basic processing unit of the encoder. During the encoding stage, the GSRM module is employed to model the interdependencies among tokens while the CSSL module extracts discriminative subspace components. In the final decision stage, the subspace components produced by the last encoder block are used to determine the class label of the central pixel in the input patch.

3.1. Spectral embedding for unified token representation

The graphical illustration of the spectral embedding part is shown in Fig. 5(a). Its main functions are feature projection and data structure transformation. In the feature projection process, a learnable linear transformation is employed to project the original high-dimensional spectral features into a latent embedding space with a predefined dimensionality, thereby enabling a unified representation of inputs with different spectral band configurations. Let Y_i denote the patch data constructed with the i th pixel as the center, and the computation can be expressed as:

$$Y'_i = \text{Norm}(\text{Linear}(Y_i)) \quad (1)$$

where $\text{Linear}(\cdot)$ denotes the linear mapping operation, which can be easily implemented using a 2D convolutional layer. $\text{Norm}(\cdot)$ represents

the batch normalization operation, used to unify the value distribution and improve training stability. Y'_i represents the patch data following feature projection.

Subsequently, during the structural transformation process, the patch size of Y'_i is adjusted through a spatial adaptive sampling (SAS) operation to achieve a unified size across inputs, after which the resulting feature representation is flattened into a token sequence for subsequent modeling. This process can be expressed as:

$$T = \text{Reshape}(\text{SAS}(Y')) \quad (2)$$

where $T = \{T_1, T_2, \dots, T_n\}$ denotes a sequence containing n tokens, and the SAS operation can be implemented using an adaptive pooling layer. Denoting the feature projection output dimension as d and the SAS-adjusted patch size as $s \times s$. Then, the sequence T consists of $n = s \times s$ tokens of dimension d . In this context, for inputs with varying patch sizes and numbers of spectral bands, the spectral embedding part standardizes them into a unified format, facilitating the encoder to perform consistent feature encoding.

3.2. Encoder of SRTF

The SRTF encoder is constructed by stacking multiple encoder blocks, each sharing the same design. As shown in Fig. 5(b), each encoder block contains two key functional modules, GSRM and CSSL, which are connected through the residual structure. In contrast to the black-box designed SAM and MLP modules in the standard Transformer encoder, our design replaces these components with GSRM and CSSL, which are grounded in the optimization principle of the representation model and the subspace decomposition process.

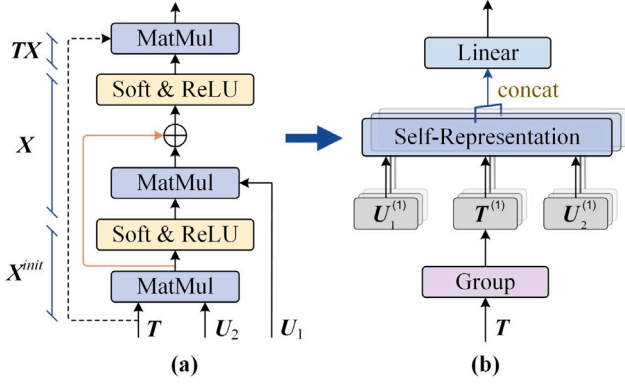


Fig. 6. Graphical illustration of SRM and GSRM. (a) SRM inference process, (b) GSRM inference process with grouping-based parallelization.

3.2.1. GSRM module for capturing interdependencies

We first introduce SRM, which serves as the foundation for constructing GSRM. The design of SRM is theoretically grounded in the optimization solution of the sparse representation model, and its core idea is to use the token sequence itself as a dictionary to adaptively capture the interdependencies among tokens.

For the sparse representation model, its mathematical formulation can be expressed as:

$$\min_X \frac{1}{2} \|DX - T\|_F^2 + \lambda \|X\|_1 + N_+(X) \quad (3)$$

where D denotes the dictionary and X represents the representation coefficients to be solved. $\|DX - T\|_F^2$ serves as the fidelity term, λ is the regularization parameter for the sparsity term $\|X\|_1$, and $N_+(\cdot)$ denotes the non-negativity constraint. By applying the iterative shrinkage-thresholding algorithm (ISTA), the solution of Eq. (3) at the k th iteration is given by two steps:

$$\begin{aligned} R^{(k)} &= (I - \eta D^T D)X^{(k-1)} + \eta D^T T, \\ X^{(k)} &= \text{ReLU}(\text{soft}_{\lambda}(R^{(k)})) \end{aligned} \quad (4)$$

where η is an update step size that needs to be set manually, $\text{soft}_{\lambda}(\cdot)$ denotes the soft-thresholding operator with λ as the threshold, and $\text{ReLU}(\cdot)$ is applied to enforce the non-negativity constraint. Subsequently, the terms $I - \eta D^T D$ and ηD^T in Eq. (4) can be replaced with two independent learnable variables (Gregor et al., 2010), denoted as U_1 and U_2 , respectively, thereby enabling fast optimization:

$$R^{(k)} = U_1 X^{(k-1)} + U_2 T \quad (5)$$

Here, the introduced learnable variables can be adaptively optimized through backpropagation, the need for solving coefficients through numerous iterative computations is effectively eliminated. Accordingly, we simplify the solution process of Eq. (3) into two essential iterations, where the first iteration is used to initialize the representation coefficients and the second iteration is used to refine and solve them:

$$\begin{aligned} X^{\text{init}} &= \text{ReLU}(\text{soft}_{\lambda}(U_2 T)), \\ X &= \text{ReLU}(\text{soft}_{\lambda}(U_1 X^{\text{init}} + U_2 T)) \end{aligned} \quad (6)$$

where X^{init} denotes the initialized representation coefficients, and λ is set as a learnable parameter. As shown in Fig. 6(a), the structure of SRM is designed strictly according to Eq. (6). Specifically, the input variables T , U_2 , and U_1 in SRM correspond directly to the variables in Eq. (6) and collectively implement the computational process defined by the equation. Notably, the SRM output is obtained by using T itself as the dictionary, producing TX . In this way, each output token is a linear combination of all input tokens, effectively capturing the critical interdependencies among tokens.

However, the proposed SRM lacks the parallel interactions capability similar to the multi-head version of self-attention, preventing it

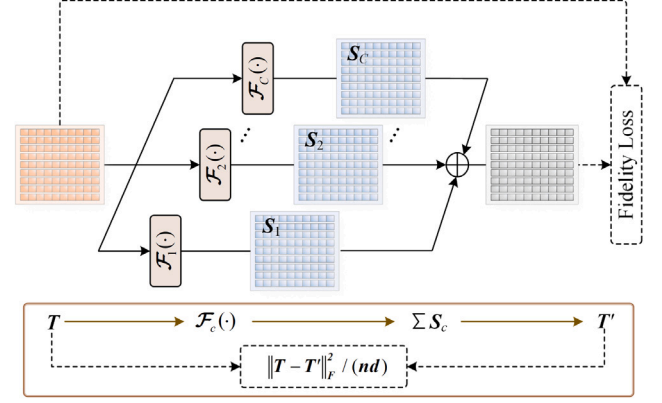


Fig. 7. Graphical illustration of the proposed CSSL module.

from capturing interdependencies among local features. Therefore, we further propose the GSRM module. As illustrated in Fig. 6(b), the token sequence T is divided into g groups, each with a dimension of p/g . The SRM module is independently applied to each group to model the internal interdependencies, and the outputs of these groups are then concatenated to reconstruct the complete representation. The overall computation process can be expressed as:

$$\begin{aligned} GSRM(T) &= \text{Concat}(\text{group}^{(1)}, \dots, \text{group}^{(g)}), \\ \text{group}^{(i)} &= SRM(U_1^{(i)}, T^{(i)}, U_2^{(i)}) \end{aligned} \quad (7)$$

where $\text{Concat}(\cdot)$ denotes the concatenation operation, and $U_1^{(i)}$, $T^{(i)}$ and $U_2^{(i)}$ represent the inputs to the i th group SRM module. Moreover, a linear mapping layer is introduced at the end to enable information interaction across groups.

3.2.2. CSSL module for extracting subspace components

In the classification process based on representation models, one of the core steps involves reconstructing subspace components for different classes using representation coefficients and the corresponding sub-dictionaries, which serve as the basis for subsequent decision-making. This process can be interpreted as decomposing the input T into a sum of subspace components, each associated with a distinct class:

$$T = S_1 + S_2 + \dots + S_c + \dots + S_C \quad (8)$$

where S_c represents the c th subspace components, and C is the number of land-cover classes contained in HRSI. Motivated by this process, we design the CSSL module to effectively extract discriminative subspace components corresponding to different classes, providing a reliable basis for downstream classification task.

As illustrated in Fig. 7, we first introduce multiple learnable functional modules, and the c th functional module is denoted as $F_c(\cdot)$. Then, the input token sequence T is processed in parallel through each functional module to extract distinct subspace components, and their sum T' is computed by element-wise addition of all components:

$$T' = F_1(T) + F_2(T) + \dots + F_c(T) + \dots + F_C(T) \quad (9)$$

Here, $F_c(T)$ corresponds to S_c in Eq. (8). For each functional module, we implement a single-hidden-layer autoencoder, where the feature dimension of the hidden layer is set to half that of the input layer, and $\text{ReLU}(\cdot)$ is utilized as the activation function. This autoencoder architecture leverages a compress-then-reconstruct mechanism to effectively learn low-dimensional subspace representations.

However, at this stage, the reconstructed output T' does not satisfy the equality constraint in Eq. (8). In other words, without ensuring that T' is equivalent to the input T , the internal decomposition mechanism

of the CSSL module lacks theoretical validity. To address this issue, a fidelity loss, denoted as $\mathcal{L}_{fidelity}$, is introduced:

$$\begin{aligned}\mathcal{L}_{fidelity} &= \frac{1}{nd} \|T - T'\|_F^2 \\ &= \frac{1}{nd} \left\| T - \sum_{c=1}^C \mathcal{F}_c(T) \right\|_F^2\end{aligned}\quad (10)$$

During network optimization, minimizing the fidelity loss is incorporated as one of the learning objectives, thereby encouraging the CSSL module to adhere to the process of subspace decomposition.

3.3. Design of loss function

In the proposed SRTF, the overall loss function consists of two terms. The first term is the fidelity loss $\mathcal{L}_{fidelity}$ introduced in the CSSL module, and its definition has been provided in Eq. (10). The second is the label loss $\mathcal{L}_{label}$, which measures the discrepancy between the predicted probabilities and the ground-truth class labels.

For the Transformer-based methods, the final result is usually determined by the high-level semantic representation encoded in the class token, making the decision process relatively opaque. In contrast, the proposed CSSL, derived from the subspace decomposition process, explicitly extracts the subspace components that contribute to decision-making. In this case, to eliminate the reliance on class token and to further enhance the transparency of the decision-making process, we convert these components into probability outputs corresponding to each class:

$$p_c = \text{Softmax}(S_c) = \frac{e^{\|S_c\|_F^2}}{\sum_{i=1}^C e^{\|S_i\|_F^2}} \quad (11)$$

where e denotes the natural constant and p_c represents the probability of predicting the input patch data as the c th class. Assuming the total number of training samples is N_t , with the corresponding class labels denoted as $\{l_1, l_2, \dots, l_{N_t}\}$, the label loss is constructed in the form of cross-entropy:

$$\mathcal{L}_{label} = -\frac{1}{N_t} \sum_{i=1}^{N_t} \sum_{c=1}^C \Phi(c, l_i) \log(p_c) \quad (12)$$

Here, Φ is an indicator function, which takes the value of 1 when c matches l_i and 0 otherwise.

Subsequently, considering that the encoder consists of K blocks, and each block has its corresponding fidelity loss, the final loss function of SRTF can be expressed as:

$$\mathcal{L}_{total} = \mathcal{L}_{label} + \frac{\alpha}{N_t K} \sum_{i=1}^{N_t} \sum_{k=1}^K \mathcal{L}_{fidelity}^{(i,k)} \quad (13)$$

where $\mathcal{L}_{fidelity}^{(i,k)}$ is the fidelity loss of the i th training sample in the k th CSSL module, and α is a weighting parameter used to balance the contributions of the two types of losses.

3.4. Training and application of the proposed SRTF

In the training phase, the number of training epochs is uniformly set to 200 for all datasets, with a batch size of 32 and a learning rate of $1e-4$. The number of groups in the GSRM module is set to 4. The balance parameter α in the loss function is empirically set to 0.01, and the optimizer is configured as Adam. Regarding the patch size, different settings are adopted to fully exploit the classification performance of the network. Specifically, the patch size is set to 27×27 for HongHu, 7×7 for Yellow River Delta, 31×31 for SZUTreeHSI, and 15×15 for Hohai University. For varying patch sizes and spectral bands, the spectral embedding module adaptively converts the input into a unified representation of $\{n = 7 \times 7, d = 64\}$. In addition, a series of data augmentation techniques, such as random rotation and flipping, are employed to enhance the diversity of training samples. After training,

the optimized SRTF model predicts the class label for each pixel, and these predictions are then combined to generate the final land cover map.

4. Experimental design

4.1. Comparison with advanced methods

To evaluate the performance of our method, seven advanced classification methods were selected for comparison, primarily focusing on the Transformer architecture. These include spectralformer (SFormer) (Hong et al., 2022), spectral-spatial Transformer network (SSTN) (Zhong et al., 2021), spectral-spatial-scale attention network (S3ANet) (Hu et al., 2022), group-aware hierarchical Transformer (GAHT) (Mei et al., 2022), center-to-surrounding interactive learning (CSIL) (Yang et al., 2023), diffusion-enhanced masked autoencoder (DEMAE) (Z. Li et al., 2024), and CNN-Transformer and channel-spatial attention network (CTANet) (Fu et al., 2025). Among these, SFormer uses pixel-wise input with a 1×1 patch size, while the other methods use patch-wise input with our settings. Section 5 presents a comprehensive comparison based on quantitative metrics and visual assessments.

4.2. Model configuration analysis

To assess the impact of different model configurations on the performance of SRTF, in Section 6.1, we conducted the parameter sensitivity analysis and further examine the performance of our method under varying levels of supervision. Specifically, Sections 6.1.1 and 6.1.2 present analyses of the performance trends of SRTF with different patch sizes and learning rate settings, respectively. These analyses aim to assess the influence of these hyperparameters and provide guidance for practical parameter selection. Furthermore, Section 6.1.3 investigates the performance of SRTF and the comparison methods with different numbers of training samples, assessing the stability of our model in different land cover scenarios with limited labeled data.

4.3. Discussion of effectiveness and limitations

To further evaluate the effectiveness of the proposed method, we conducted several analyses in Section 6.2. First, Section 6.2.1 presents ablation studies to quantitatively evaluate the contribution of each SRTF component to overall performance. Next, Sections 6.2.2 and 6.2.3 provide in-depth analyses and validations of the effectiveness of the two key modules, SRM and CSSL, respectively. Finally, Section 6.2.4 evaluates the convergence of the loss function by analyzing trends in the training loss. Furthermore, Section 6.3 summarizes the current limitations and outlines potential directions for future research.

5. Land cover mapping results

Performance of each method in land cover mapping is evaluated using multiple metrics, including per-class accuracy, overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). Classification maps generated by each method are also presented for visual assessment. It is worth noting that all reported metric values are averaged over 10 independent runs, with displayed maps corresponding to results closest to the average performance. For the HongHu, SZUTreeHSI, and Hohai University datasets, which contain sufficient labeled samples, 50 samples per class are randomly selected for training, and the remaining labeled samples are used for testing. Due to the limited number of labeled samples in the Yellow River Delta dataset, particularly for the *Aquatic vegetation* class with only 39 samples, 20 samples per class are used for training. Consequently, the training samples account for approximately 0.25%, 6.5%, 0.06%, and 0.06% of the total labeled samples in the HongHu, Yellow River Delta, SZUTreeHSI, and Hohai University datasets, respectively.

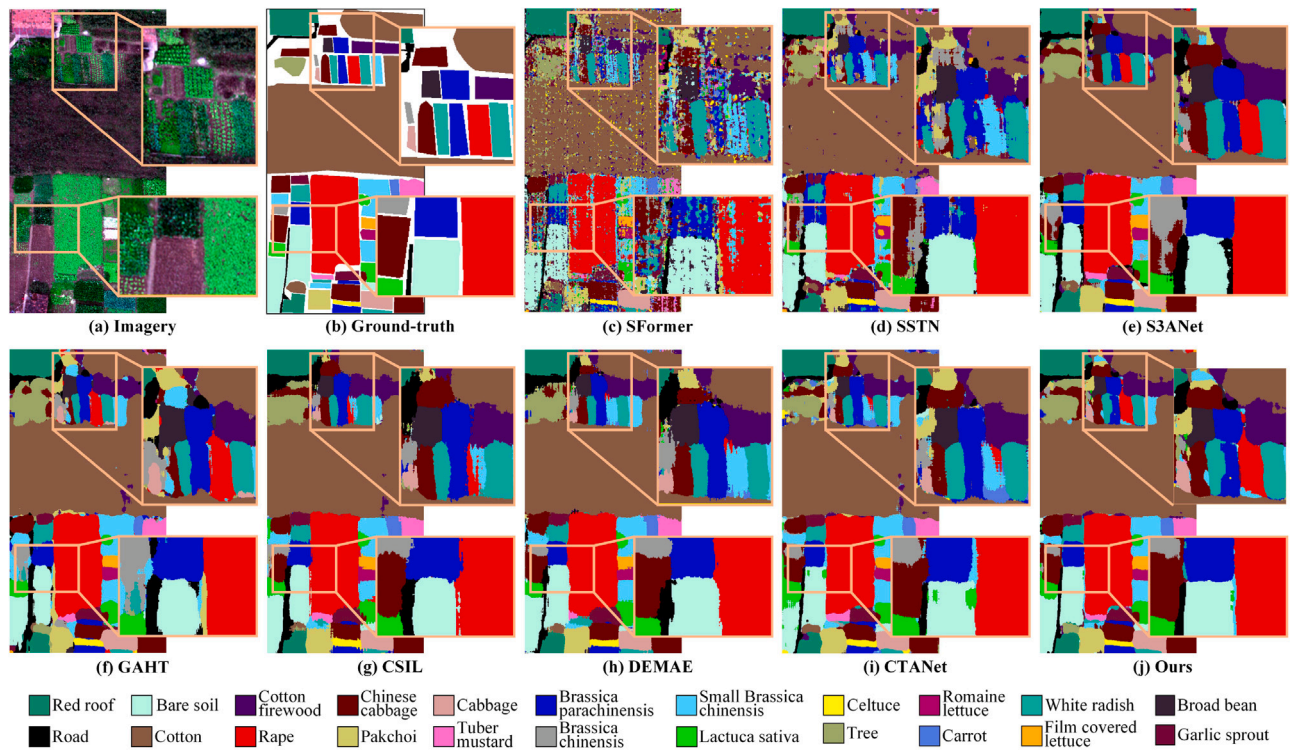


Fig. 8. Classification maps produced by different methods for the Honghu dataset.

Table 2

Classification accuracies for Honghu, with the best and second-best results highlighted in bold and underlined.

No.	Class name	Training	SFormer	SSTN	S3ANet	GAHT	CSIL	DEMAE	CTANet	Ours
1	Red roof	50	89.48±0.90	86.59±5.23	<u>96.22±1.94</u>	95.80±1.33	92.45±2.90	93.52±4.08	96.73±2.23	95.65±2.13
2	Road	50	76.62±3.34	76.08±4.02	93.39±2.56	90.88±4.38	97.31±1.70	91.51±6.12	<u>96.84±2.19</u>	96.59±1.38
3	Bare soil	50	78.60±0.82	82.91±3.13	<u>92.90±3.06</u>	82.70±5.87	87.85±3.61	92.04±1.52	<u>91.27±3.46</u>	96.39±0.58
4	Cotton	50	79.51±4.13	97.43±0.74	97.16±1.52	97.34±1.38	97.75±0.63	<u>98.21±0.81</u>	97.83±0.78	98.57±0.46
5	Cotton firewood	50	78.43±3.87	95.77±3.74	98.86±0.67	95.91±4.23	<u>99.51±0.39</u>	99.56±0.59	99.41±0.36	99.23±0.78
6	Rape	50	86.58±1.46	89.93±3.34	94.25±1.41	95.14±2.17	97.41±1.65	<u>96.22±0.88</u>	93.10±2.26	95.83±1.90
7	Chinese cabbage	50	59.43±3.17	67.77±6.88	88.34±3.66	74.60±8.78	89.06±2.11	89.30±0.51	<u>90.43±3.40</u>	91.46±1.19
8	Pakchoi	50	61.62±5.82	94.51±1.75	96.51±2.33	96.21±2.77	97.69±2.28	99.90±0.19	98.21±1.31	<u>99.50±0.40</u>
9	Cabbage	50	92.05±1.30	93.65±1.64	97.47±1.52	93.93±2.54	97.09±1.39	97.93±0.68	96.31±2.09	<u>97.76±1.83</u>
10	Tuber mustard	50	57.04±4.04	81.08±1.44	<u>92.95±4.12</u>	87.82±3.12	91.82±3.05	92.60±1.48	93.64±3.46	<u>92.61±1.45</u>
11	Brassica parachinensis	50	56.32±3.92	74.87±7.37	91.63±3.12	89.15±3.83	90.11±2.20	92.57±1.84	90.81±5.84	94.96±1.22
12	Brassica chinensis	50	44.31±3.68	66.86±5.36	88.54±3.56	86.81±4.06	94.58±3.17	95.20±2.84	<u>96.03±1.34</u>	97.51±2.21
13	Small Brassica chinensis	50	51.57±5.94	69.08±2.88	85.34±5.10	78.41±10.06	85.46±6.29	<u>88.74±3.14</u>	87.28±7.37	90.17±2.42
14	Lactuca sativa	50	64.65±7.30	90.23±2.91	95.11±1.92	97.22±1.81	96.72±1.14	91.95±5.40	<u>97.25±2.18</u>	97.75±1.10
15	Celtuce	50	91.95±1.07	98.09±2.15	99.38±0.82	99.69±0.32	98.81±1.25	99.50±0.74	100.0±0.00	<u>99.85±0.25</u>
16	Film covered lettuce	50	78.27±1.32	91.44±2.79	<u>98.49±2.05</u>	98.42±2.65	95.30±2.90	96.83±1.71	97.87±0.14	98.83±1.28
17	Romaine lettuce	50	79.73±5.25	99.69±0.33	98.37±2.25	94.57±4.91	99.47±0.40	99.74±0.21	<u>99.71±0.32</u>	99.57±0.53
18	Carrot	50	91.89±2.47	98.05±1.46	97.01±2.68	97.45±4.37	98.12±1.43	99.33±0.67	99.75±0.22	98.60±0.80
19	White radish	50	81.55±2.80	81.35±4.63	92.68±2.86	88.29±5.02	93.81±3.49	92.14±2.30	<u>93.99±3.51</u>	94.00±2.72
20	Garlic sprout	50	64.56±5.25	84.12±3.56	97.77±0.73	95.93±3.94	98.82±1.04	98.81±1.05	<u>97.72±1.80</u>	98.29±1.12
21	Broad bean	50	83.82±3.37	99.60±0.38	99.91±0.19	98.09±2.77	99.92±0.10	<u>99.97±0.06</u>	100.0±0.00	<u>100.0±0.00</u>
22	Tree	50	78.52±3.63	94.24±2.63	98.52±2.76	<u>99.78±0.35</u>	99.21±0.76	99.49±0.38	98.31±1.94	99.95±0.08
Overall accuracy (%)			75.43±1.73	88.90±0.54	94.74±0.70	92.53±1.08	95.12±0.59	<u>95.69±0.50</u>	95.35±0.81	96.67±0.38
Average accuracy (%)			73.93±0.90	86.97±0.60	95.04±0.49	92.46±0.68	95.38±0.65	95.68±0.42	<u>96.02±0.47</u>	96.96±0.31
Kappa coefficient (×100)			70.30±1.81	86.05±0.65	93.40±0.86	90.63±1.32	93.86±0.74	<u>94.57±0.56</u>	94.16±1.01	95.80±0.43

5.1. Results in the agricultural scenario

The detailed classification results of each method on the HongHu dataset, which represents an agricultural scenario, are presented in Table 2, and the corresponding classification evaluation maps are shown in Fig. 8. In terms of the three comprehensive evaluation metrics, the proposed method achieves improvements of 0.98%–21.24% in OA, 0.98%–23.03% in AA, and 1.23%–25.5% in κ over the comparison methods, demonstrating significant performance advantages. Regarding per-class accuracy, although methods such as CSIL, DEMAE, and CTANet demonstrate superior performance in certain specific classes, the proposed

method achieves overall better classification performance. Among the 22 classes in the dataset, our method attains the highest accuracy in 11 classes, with particularly notable improvements for *Bare soil* and *Small Brassica chinensis*.

From the classification maps, it is evident that SFormer, which relies solely on spectral information, is easily affected by the similar spectral characteristics of different crops. This leads to a large number of misclassified pixels resembling salt-and-pepper noise in the resulting maps, indicating poor spatial consistency. In contrast, the other methods that integrate both spectral and spatial information produce

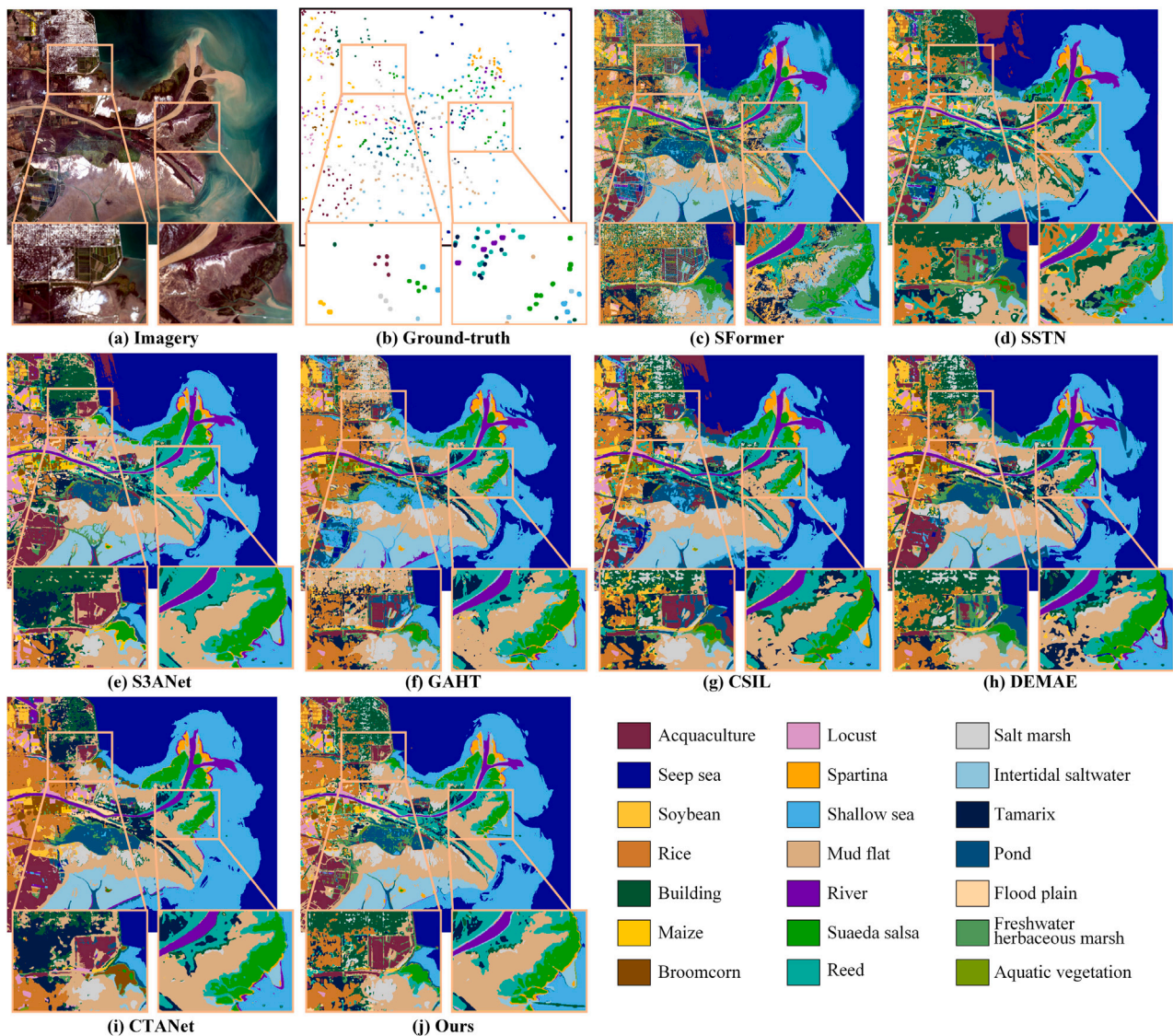


Fig. 9. Classification maps produced by different methods for the Yellow River Delta dataset.

visually smoother maps with better spatial continuity. Furthermore, the subfigures reveal that the performance differences among methods significantly affect the quality of maps. For example, compared with other patch-wise based methods, SSTN suffers from more pronounced boundary blurring between different crops, while GAHT exhibits noticeable over-smoothing effects. In contrast, the map produced by the proposed method exhibit clearer boundaries, better smoothness, and higher accuracy, fully validating its effectiveness in this scene.

5.2. Results in the wetland scenario

Table 3 summarizes the quantitative classification results on the Yellow River Delta dataset, which contains diverse land cover types across a complex wetland environment. Fig. 9 shows the corresponding classification maps. The proposed method consistently outperforms the comparison methods, with gains ranging from 3.74% to 13.03% in OA, 2.22% to 10.37% in AA, and 4.06% to 14.07% in κ . From the perspective of per-class accuracy, the proposed method delivers the highest performance in over half of the classes, achieving perfect accuracy (100%) in seven of them. Particularly, it demonstrates substantial superiority over the comparison methods in *Building* and *Shallow sea*.

From the visualized classification maps, the proposed method demonstrates higher overall consistency and continuity. The locally enlarged subfigures further show that, compared with patch-wise methods such as SSTN, GAHT, and DEMA, it better preserves the morphological characteristics of ground objects and produces clearer boundaries. For the pixel-wise SFormer, the absence of spatial smoothing allows relatively clear depiction of small object boundaries, but its overall mapping accuracy remains significantly lower than that of the other methods. Overall, the proposed method achieves a superior balance between boundary detail preservation and global classification accuracy, highlighting its potential for fine-scale wetland mapping applications.

5.3. Results in the forestry scenario

As shown in Table 4, for the classification results on the SZUTreeHSI dataset, the proposed method outperforms comparison methods across all evaluation metrics. Among the 20 tree species in this dataset, our method achieves the best performance in 8 tree species, with particularly notable advantages in *Roystonea regia* and *Delonix regia*. Meanwhile, other comparison methods also show competitive performance, with DEMA achieving the best results in 6 tree species and CTANet in 3 tree species. Furthermore, our method attains the

Table 3

Classification accuracies for Yellow River Delta, with the best and second-best results highlighted in bold and underlined.

No.	Class name	Training	SFormer	SSTN	S3ANet	GAHT	CSIL	DEMAE	CTANet	Ours
1	Acquaculture	20	79.95±5.23	72.48±10.41	92.39±2.11	87.47±10.15	73.19±5.84	88.68±1.35	88.53±2.31	<u>91.15±2.96</u>
2	Seep sea	20	90.53±8.37	97.94±1.46	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
3	Soybean	20	98.00±0.89	98.00±1.26	99.11±0.83	85.40±16.72	99.33±0.54	<u>99.31±0.57</u>	99.11±1.30	92.44±1.56
4	Rice	20	82.67±5.11	86.78±1.77	86.00±5.07	88.22±0.65	88.59±1.42	86.82±4.33	87.41±0.29	89.41±1.05
5	Building	20	47.95±7.09	88.49±5.59	77.14±9.29	92.05±4.70	91.75±3.94	87.24±13.00	89.84±2.94	96.51±4.96
6	Maize	20	98.37±1.58	91.63±3.15	96.84±5.10	55.35±26.30	99.74±0.53	79.47±17.31	99.21±1.58	100.0±0.00
7	Broomcorn	20	94.12±5.81	95.76±3.21	98.67±2.67	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
8	Locust	20	100.0±0.00	99.40±0.80	98.53±2.68	85.37±19.25	99.69±0.63	98.32±2.05	93.40±4.95	97.91±3.69
9	Spartina	20	94.63±2.22	93.68±0.00	100.0±0.00	99.58±0.84	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
10	Shallow sea	20	57.43±21.11	79.14±7.52	<u>83.01±14.83</u>	70.56±14.97	79.24±10.27	73.62±10.59	81.16±9.44	97.99±1.27
11	Mud flat	20	96.17±2.62	97.27±2.84	99.29±0.60	99.89±0.22	99.89±0.23	99.09±1.37	98.35±3.30	99.77±0.36
12	River	20	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00
13	Suaeda salsa	20	91.05±7.52	96.13±1.89	<u>99.65±0.70</u>	91.00±6.01	100.0±0.00	97.89±4.22	98.89±2.23	100.0±0.00
14	Reed	20	77.65±5.43	78.17±6.88	67.55±3.81	73.57±6.54	79.64±8.75	<u>80.33±10.23</u>	79.36±8.94	80.64±7.47
15	Salt marsh	20	99.93±0.14	99.21±1.15	93.74±4.06	98.91±2.19	<u>99.86±0.13</u>	99.72±0.39	92.52±3.50	99.48±0.50
16	Intertidal saltwater	20	96.71±2.49	95.27±3.16	97.70±3.81	98.83±1.05	94.88±3.42	97.82±1.81	99.68±0.18	99.68±0.54
17	Tamarix	20	75.61±4.27	74.96±8.41	87.61±15.52	<u>95.61±4.92</u>	91.86±5.77	96.96±2.04	92.57±2.99	95.58±3.58
18	Pond	20	49.48±23.76	38.53±10.84	82.46±12.55	57.00±15.04	49.64±13.20	84.88±12.33	97.93±2.36	95.07±3.64
19	Flood plain	20	93.45±2.01	82.07±7.91	99.17±1.67	92.76±5.27	91.67±7.68	74.87±15.71	91.67±3.49	<u>97.08±4.86</u>
20	Freshwater herbaceous marsh	20	90.65±4.38	79.68±6.09	<u>97.31±1.54</u>	98.06±3.87	81.15±14.51	96.54±2.83	93.85±4.62	<u>97.31±4.49</u>
21	Aquatic vegetation	20	<u>97.93±1.69</u>	93.10±3.78	100.0±0.00	100.0±0.00	97.89±4.21	100.0±0.00	100.0±0.00	100.0±0.00
Overall accuracy (%)			84.39±3.11	87.88±1.41	92.89±1.99	88.76±2.45	90.29±1.11	92.18±1.25	93.68±1.33	97.42±0.37
Average accuracy (%)			86.30±1.05	87.51±0.65	93.15±1.79	89.03±4.38	91.33±1.39	92.46±2.53	<u>94.45±0.45</u>	96.67±0.35
Kappa coefficient (×100)			83.12±3.28	86.83±1.51	92.28±2.14	87.82±2.62	89.42±1.20	91.50±1.35	<u>93.13±1.43</u>	97.19±0.40

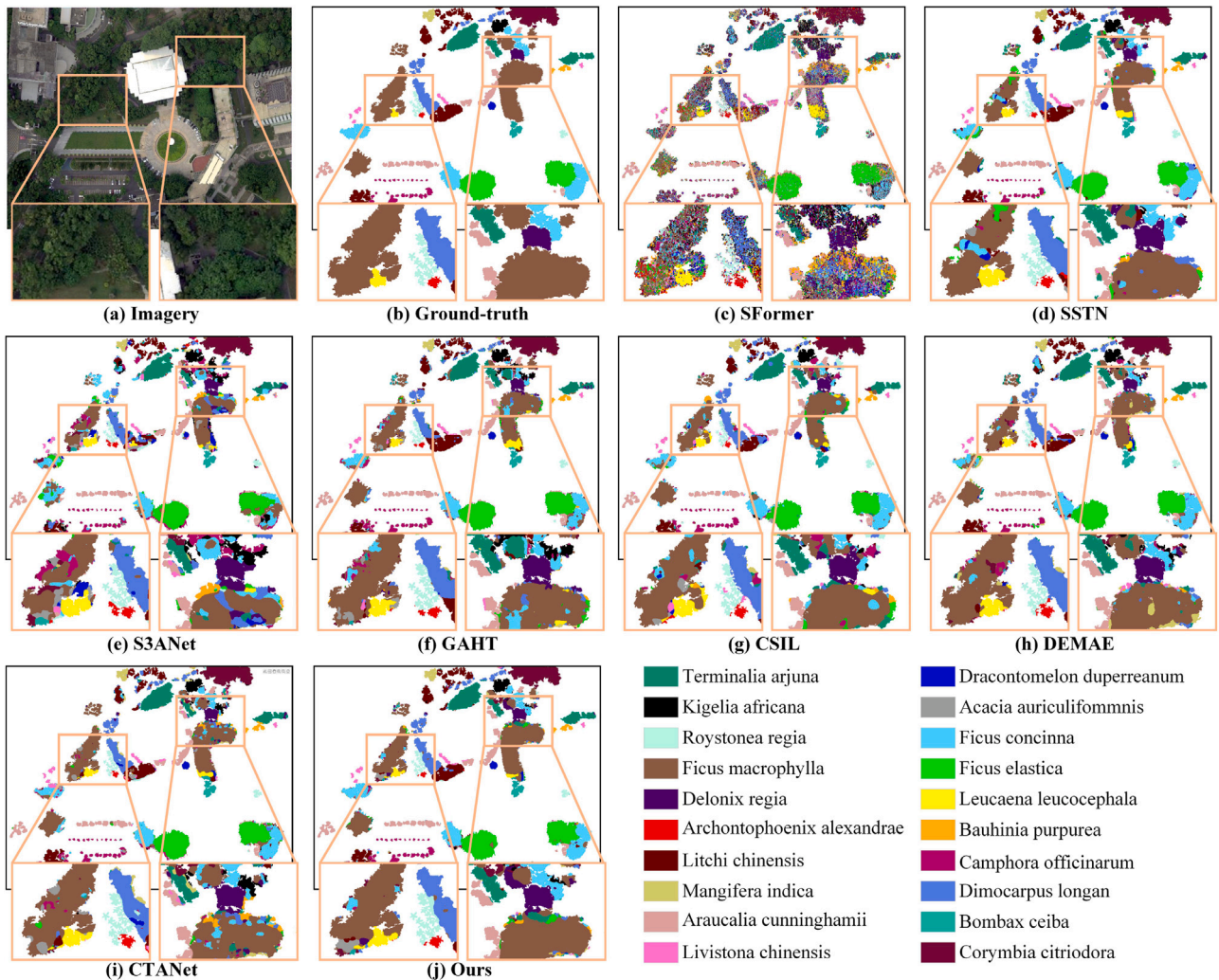
**Fig. 10.** Classification maps produced by different methods for the SZUTreeHSI dataset.

Table 4

Classification accuracies for SZUTreeHSI, with the best and second-best results highlighted in bold and underlined.

No.	Class name	Training	SFormer	SSTN	S3ANet	GAHT	CSIL	DEMAE	CTANet	Ours
1	Terminalia arjuna	50	62.17±3.52	80.95±3.90	<u>93.07±3.11</u>	93.01±2.40	91.72±2.66	92.11±4.65	91.04±4.09	93.32±3.36
2	Kigelia africana	50	69.19±5.44	93.04±1.49	<u>98.42±1.14</u>	94.49±3.90	<u>99.67±0.43</u>	99.86±0.15	97.87±1.60	99.09±0.83
3	Roystonea regia	50	39.36±2.86	85.84±1.77	<u>95.02±3.57</u>	92.25±1.67	94.64±3.59	95.01±1.51	92.84±3.24	96.35±1.40
4	Ficus macrophylla	50	19.53±3.70	46.25±6.03	<u>76.94±2.88</u>	67.35±3.72	66.34±3.38	70.77±3.47	70.55±5.02	77.42±4.67
5	Delonix regia	50	43.73±8.33	82.98±6.31	93.88±3.89	92.20±4.93	93.92±2.48	93.84±3.32	<u>96.22±2.80</u>	97.78±0.87
6	Archontophoenix alexandrae	50	90.00±3.50	98.91±0.85	99.65±0.38	98.21±1.68	99.71±0.58	99.54±0.08	<u>99.80±0.26</u>	99.87±0.12
7	Litchi chinensis	50	36.36±4.87	59.15±5.25	<u>84.34±3.26</u>	77.87±2.15	75.10±5.10	84.71±6.49	78.08±7.79	82.18±5.61
8	Mangifera indica	50	33.01±2.68	86.18±7.63	<u>94.83±1.17</u>	95.69±1.14	98.10±1.95	<u>98.17±1.64</u>	98.33±1.63	97.54±0.76
9	Araucalia cunninghamii	50	71.97±3.49	90.71±1.97	94.47±0.78	94.93±1.71	92.96±1.57	94.83±2.57	93.76±2.31	<u>94.87±1.78</u>
10	Livistonia chinensis	50	36.17±1.64	88.24±6.19	96.71±5.81	93.75±6.36	95.94±3.45	<u>99.01±0.90</u>	98.32±1.87	99.17±0.63
11	Dracontomelon duperreanum	50	42.03±7.23	72.00±4.53	95.46±2.63	91.30±1.57	91.62±2.85	90.14±5.49	92.71±1.91	<u>94.30±1.07</u>
12	Acacia auriculiformis	50	41.59±5.26	79.84±4.11	<u>93.18±1.99</u>	91.90±2.05	87.84±5.11	92.47±2.56	88.33±3.05	94.39±2.65
13	Ficus concinna	50	18.75±3.82	38.42±6.76	<u>74.69±4.60</u>	74.88±5.12	74.88±4.33	79.22±2.84	76.29±1.36	<u>76.47±4.70</u>
14	Ficus elastica	50	71.47±2.40	88.08±2.64	91.38±4.80	<u>94.09±1.05</u>	90.51±3.28	93.03±1.25	94.41±0.87	93.17±1.78
15	Leucaena leucocephala	50	81.07±5.42	99.88±0.12	100.0±0.00	99.34±0.96	99.92±0.16	100.0±0.00	99.53±0.57	<u>99.94±0.13</u>
16	Bauhinia purpurea	50	83.65±2.34	96.42±1.25	99.58±0.84	<u>99.84±0.21</u>	99.98±0.04	99.58±0.83	99.72±0.56	98.56±2.76
17	Camphora officinarum	50	47.60±3.47	79.64±6.03	88.35±7.18	88.41±4.66	94.88±1.61	95.70±2.38	94.05±1.04	<u>95.55±1.65</u>
18	Dimocarpus longan	50	30.97±4.53	59.04±7.21	81.22±9.85	77.19±5.76	84.35±4.07	83.15±6.38	88.39±4.54	<u>84.49±4.43</u>
19	Bombax ceiba	50	76.98±3.44	<u>99.87±0.15</u>	99.86±0.13	98.90±0.92	99.85±0.17	99.81±0.38	99.88±0.09	99.88±0.15
20	Corymbia citriodora	50	79.47±2.75	<u>97.22±1.10</u>	96.62±1.78	98.67±0.68	98.51±0.83	98.99±0.72	<u>98.83±0.58</u>	98.08±0.93
Overall accuracy (%)			41.40±0.73	66.10±1.32	<u>85.47±1.02</u>	82.31±0.82	81.70±1.25	84.87±1.48	83.93±1.20	86.45±0.86
Average accuracy (%)			53.75±0.58	81.13±1.03	92.38±0.90	90.71±1.14	91.52±0.35	<u>93.00±0.79</u>	92.45±0.44	93.62±0.63
Kappa coefficient (×100)			37.13±0.66	62.78±1.36	<u>83.70±1.13</u>	80.24±0.90	79.61±1.34	83.11±1.61	82.04±1.27	84.79±0.95

Table 5

Classification accuracies for Hohai University, with the best and second-best results highlighted in bold and underlined.

No.	Class name	Training	SFormer	SSTN	S3ANet	GAHT	CSIL	DEMAE	CTANet	Ours
1	Grassland	50	78.96±2.79	87.73±1.98	<u>90.57±4.23</u>	88.72±1.69	88.03±4.04	89.64±3.72	87.31±3.43	91.44±3.17
2	Tree/shrub	50	<u>83.75±2.61</u>	66.64±5.05	74.33±7.62	71.71±4.17	80.24±3.16	79.15±2.64	88.35±2.68	81.86±4.09
3	Roof type-1	50	87.88±2.50	91.72±2.69	94.40±3.64	96.21±0.57	96.28±1.06	97.11±1.28	96.57±1.72	96.16±1.88
4	Roof type-2	50	99.04±0.79	99.53±0.54	99.08±0.66	<u>99.73±0.12</u>	99.26±0.21	99.94±0.05	99.08±0.26	99.34±0.44
5	Roof type-3	50	96.71±1.36	<u>99.76±0.17</u>	99.49±0.19	99.27±0.44	99.42±0.18	99.41±0.12	99.25±0.20	99.79±0.18
6	Roof type-4	50	99.66±0.10	99.87±0.04	<u>99.92±0.14</u>	99.67±0.13	99.88±0.12	99.69±0.01	99.87±0.12	99.95±0.16
7	Glass roof	50	99.07±0.21	99.80±0.10	99.98±0.02	99.48±0.22	99.87±0.14	99.90±0.09	<u>99.94±0.03</u>	99.83±0.16
8	Plastic mesh	50	84.95±4.13	97.92±0.73	99.49±0.56	<u>99.83±0.33</u>	99.23±0.35	99.92±0.09	99.36±0.74	98.40±3.21
9	Photovoltaic panel	50	99.53±0.31	<u>99.93±0.08</u>	99.29±0.85	99.95±0.02	99.66±0.28	99.74±0.33	99.74±0.21	99.70±0.68
10	Artificial turf	50	<u>99.98±0.01</u>	99.74±0.25	99.74±0.25	99.78±0.25	99.90±0.08	99.99±0.01	99.93±0.09	99.77±0.23
11	Stadium seat	50	96.30±1.23	98.79±1.29	99.72±0.08	99.64±0.46	<u>99.89±0.10</u>	99.92±0.12	99.48±0.51	99.32±0.63
12	Sports field-1	50	88.48±1.73	77.81±3.31	94.48±2.69	87.51±4.82	85.54±3.21	89.83±2.85	96.28±2.37	96.11±2.55
13	Sports field-2	50	71.80±3.16	73.07±6.16	87.20±3.46	86.07±3.74	81.08±3.28	<u>89.59±3.74</u>	88.82±5.87	89.71±2.83
14	Ceramic tile	50	98.49±1.34	99.73±0.26	98.62±0.71	99.40±0.54	<u>99.50±0.30</u>	97.56±2.95	98.87±0.79	94.20±7.48
15	Bare/sparse grass	50	92.58±3.79	95.27±1.08	98.71±0.64	97.10±1.26	97.12±1.19	97.40±1.40	97.77±1.43	<u>98.05±0.73</u>
16	Asphalt road	50	89.94±1.97	91.63±1.10	96.01±0.65	94.98±2.31	95.71±1.56	95.99±1.33	<u>97.51±1.65</u>	97.95±0.80
17	Painted plank	50	96.88±1.07	97.83±0.94	98.40±1.43	<u>99.40±0.32</u>	99.15±0.52	99.48±0.21	98.63±0.89	98.70±0.77
18	Slab/concrete	50	68.29±4.77	75.97±1.81	84.96±3.30	80.27±1.10	81.23±3.22	83.73±3.03	<u>86.01±2.35</u>	90.93±1.88
19	Waterbody	50	99.39±0.55	99.24±0.17	99.28±0.73	<u>99.73±0.21</u>	99.07±0.47	99.82±0.05	98.32±0.73	99.20±1.10
20	Rubber track	50	84.27±3.93	94.98±2.52	96.22±2.16	92.91±6.75	94.83±2.81	<u>95.27±4.81</u>	90.78±8.65	93.84±4.93
Overall accuracy (%)			84.60±1.31	87.44±0.61	92.39±0.74	90.71±0.34	91.16±0.97	92.44±0.75	93.55±0.47	94.81±0.69
Average accuracy (%)			90.80±0.72	92.35±0.13	95.49±0.32	94.57±0.21	94.74±0.19	95.65±0.31	<u>96.09±0.22</u>	96.21±0.74
Kappa coefficient (×100)			82.55±1.45	85.69±0.69	91.29±0.83	89.39±0.38	89.90±1.08	91.34±0.85	<u>92.60±0.53</u>	94.03±0.79

best results on the three comprehensive evaluation metrics, with improvements of at least 0.98% in OA, 0.62% in AA, and 1.09% in κ .

Fig. 10 presents the classification maps produced by different methods for visual analysis. Since the dataset contains some building and grassland regions, assigning them to a specific tree species is unreasonable and hinder a fair comparison of methods. Therefore, only the labeled regions are displayed in the maps. From the map produced by SFormer, the pixels of different classes are blended with each other, indicating that relying solely on spectral information is insufficient for effectively distinguishing between tree species. For other methods that incorporate both spectral and spatial information, the generated classification maps show considerable improvements. However, the subfigures reveal evident misclassifications, particularly along the boundaries between *Ficus macrophylla* and other tree species. Moreover, some comparison methods, such as SSTN and S3ANet, also suffer from extensive misclassifications within this class. In contrast, the map produced by our method is closer to the ground truth, which highlights its reliability for tree species identification.

5.4. Results in the built-up area scenario

For the Hohai University dataset, which represents built-up environments, the detailed classification results of all methods are listed in Table 5. The proposed method demonstrates superior performance, with OA improving by 1.26% to 10.21%, AA by 0.12% to 5.41%, and κ by 1.43% to 11.48%. In terms of per-class accuracy, it is interesting to note that DEMAE achieves the best results in 7 classes, while the proposed method ranks highest in 6 classes. However, for some classes, the difference between the two methods is minimal. For instance, in the *Artificial turf* class, their accuracies differ by only 0.2%.

From another perspective, the artificial facilities in this scene exhibit good regularity and clear boundaries, making the classification maps in Fig. 11 effective for visually comparing the mapping performance. A closer look at the spatial details in the subfigures reveals that the proposed method clearly outperforms the others. In the left subfigures, most comparison methods suffer from severe over-smoothing and incorrectly classify the *Slab/concrete* class as *Ceramic tile*. Although

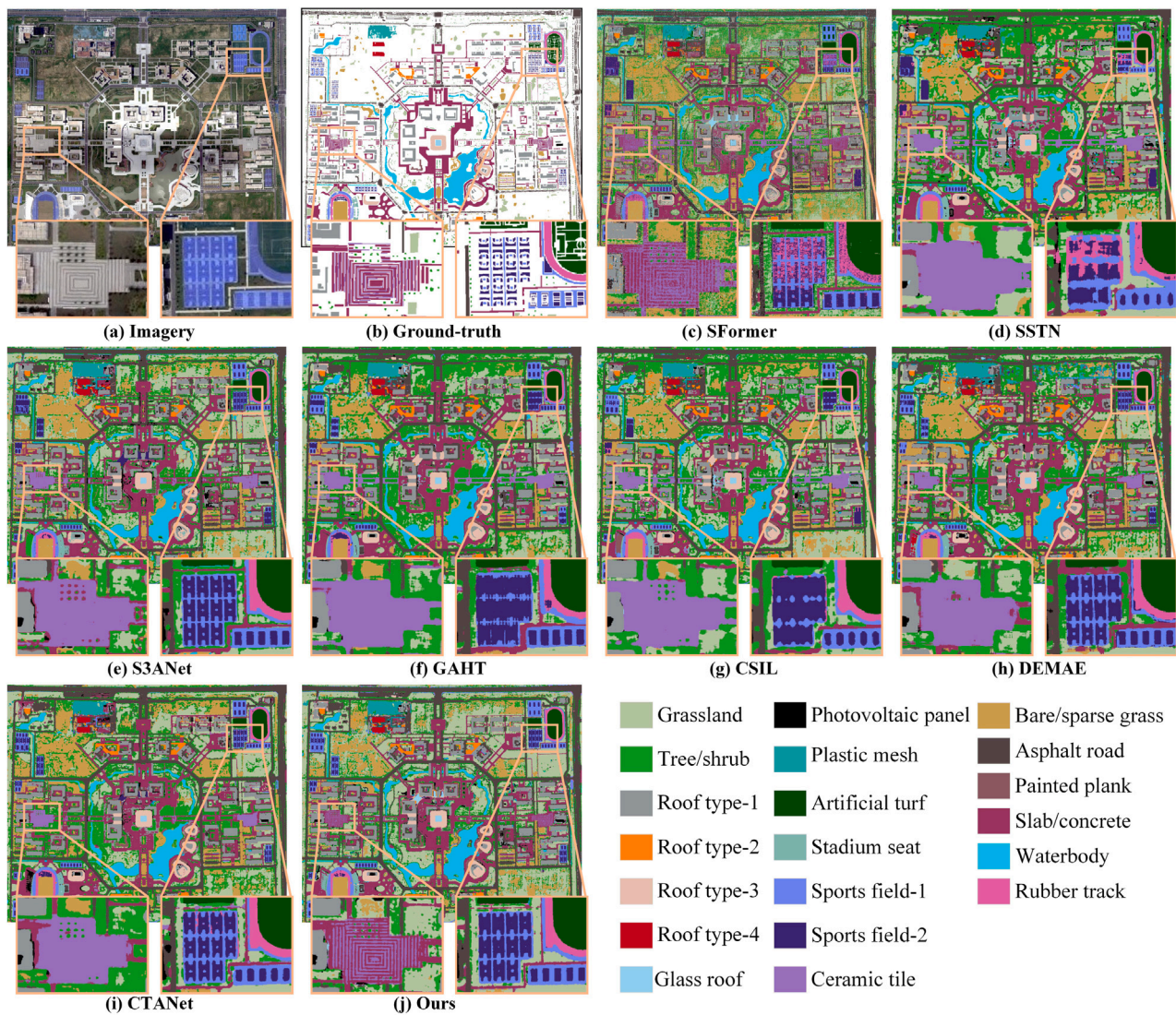


Fig. 11. Classification maps produced by different methods for the Hohai University dataset.

SFormer can distinguish the two classes, its results still show obvious salt-and-pepper noise. From the right subfigures, SSTN, GAHT, and GSIL perform noticeably worse than the other patch-wise methods, while S3ANet and our method perform comparably in this local region. However, when considering the overall performance, our method still significantly outperforms S3ANet.

6. Analysis and discussion

6.1. How model configuration affects performance

6.1.1. Effect of patch size

For HRSI, the joint utilization of spatial and spectral information is crucial for fine mapping. By tuning the patch size, the contextual range perceived by the model can be modified, which significantly influences the final classification performance. Fig. 12 illustrates the classification performance of the proposed method under different patch size settings. As the patch size increases from 3×3 to 31×31 , the classification accuracy exhibits distinct trends across the datasets. For the Honghu and Hohai University datasets, accuracy first increases and then decreases, but with different optimal sizes. Specifically, the Honghu dataset reaches the highest at 27×27 , while the Hohai University dataset at 15×15 . In contrast, the trends for the Yellow

River Delta and SZUTreeHSI datasets are opposite. For the Yellow River Delta dataset, accuracy declines with increasing patch size, while it rises steadily for SZUTreeHSI. These results demonstrate that patch size plays a critical role in classification performance, since different datasets exhibit substantial variations in spatial resolution, scene complexity, and contextual dependency. Consequently, selecting an appropriate patch size for each specific scene is essential to effectively capture discriminative spatial-spectral features and fully exploit contextual information.

6.1.2. Effect of learning rate

During the training phase, the learning rate serves as a key hyperparameter that controls the magnitude of parameter updates. An appropriate learning rate helps improve training stability and accelerates model convergence. Fig. 13 shows the impact of different learning rate settings on classification performance across four datasets. In the experiments, the learning rate ranges from $5e-6$ to $5e-3$, while the number of training epochs is fixed at 200. In the early stage, the results indicate that as the learning rate increases, the accuracy gradually improves across four datasets. This is because small learning rates lead to limited parameter updates, whereas increasing the learning rate allows the model to approach the optimal solution more quickly. Subsequently, with a further increase, the accuracy drops sharply.

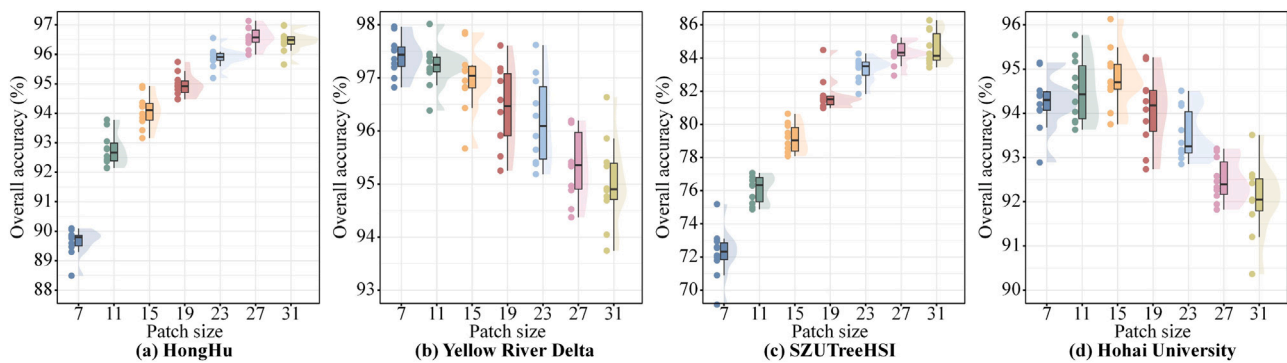


Fig. 12. Effect of different patch size settings on classification performance.

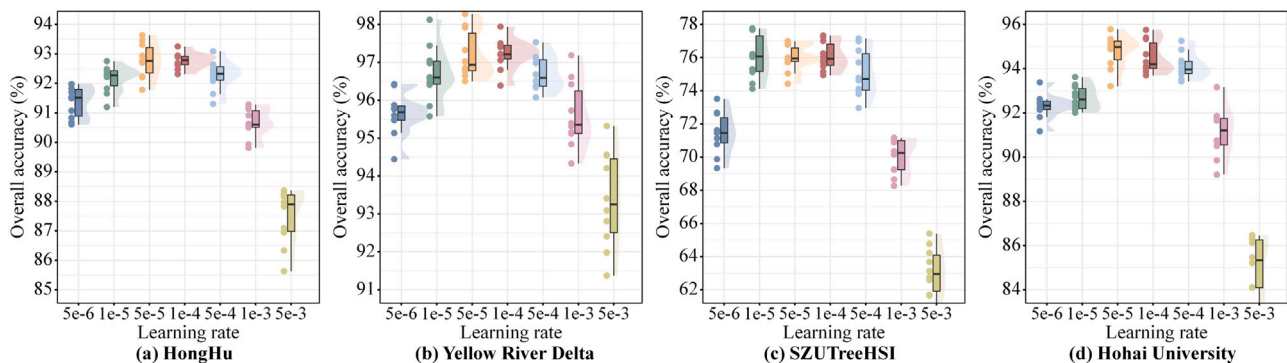


Fig. 13. Effect of different learning rate settings on classification performance.

This is because once the learning rate exceeds an appropriate range, the training process becomes unstable, ultimately causing a significant performance decline. The results also show that a learning rate of $1e-4$ achieves the best accuracy on the Honghu, Yellow River Delta, and SZUTreeHSI datasets. For the Hohai University dataset, the best accuracy occurs at $5e-5$, but the difference compared to $1e-4$ is minimal. Overall, a learning rate of $1e-4$ can be considered an optimal setting for our method.

6.1.3. Effect of different supervision levels

To further assess the superiority and generalization ability of the proposed method, we compare the classification performance of different methods under varying levels of supervision by controlling the number of training samples. Fig. 14 illustrates the changes in OA as the number of training samples per class increases from 10 to 80. The results indicate that accuracy improves for all methods as the sample size grows, but the rate of improvement varies. The most significant gains occur when the number of samples increases from 10 to 30, followed by a slower improvement between 30 and 60 samples. From 60 to 80 samples, accuracy generally stabilizes across all datasets except SZUTreeHSI. The advantage of the proposed method is particularly pronounced when training samples are limited. For example, on the Yellow River Delta dataset, with only 10 samples per class, our method achieves significantly higher accuracy than others. In contrast, SFormer and SSTN consistently underperform across all four datasets, highlighting their limitations, whereas CTANet and DEMAe exhibit more competitive performance. Furthermore, the subfigures clearly show the performance differences among the methods in the range of 40–80 samples, with the proposed method consistently maintaining the lead. These results collectively demonstrate the effectiveness of our method under varying levels of supervision.

6.2. How each component contributes to performance

6.2.1. Ablation study for quantitative evaluation

In this section, an ablation study is conducted to quantitatively evaluate the contribution of each module in SRTF to the overall classification performance. Specifically, different SRTF variants are constructed by removing or replacing certain functional modules for comparative analysis. These variants include removing the spectral embedding module (variant model-1, VM-1), removing the fidelity loss (variant model-2, VM-2), replacing the GSRM module with the multi-head self-attention (MHSA) module (variant model-3, VM-3), replacing the CSSL module with the MLP module (variant model-4, VM-4), and simultaneously replacing both the GSRM and CSSL modules (variant model-4, VM-5).

From Fig. 15, among the five variants, VM-1 shows the most significant performance degradation. This is because the spectral embedding module plays a fundamental role in reducing the redundancy of spectral information. Once it is removed, it undermines the effectiveness of subsequent modules and ultimately results in a significant decline in overall performance. The results of VM-3 and VM-4 also show a noticeable decline, indicating that the proposed GSRM and CSSL modules provide superior classification capability compared with MHSA and MLP. Meanwhile, the results of VM-5 demonstrate a certain degree of synergy between GSRM and CSSL, as simultaneously replacing both modules leads to a larger drop in accuracy. Additionally, the results of VM-2 indicate that removing the fidelity loss also affects performance, highlighting its essential role in guiding the network optimization. Overall, the ablation study clearly demonstrates the effectiveness of each module in SRTF for enhancing classification capability.

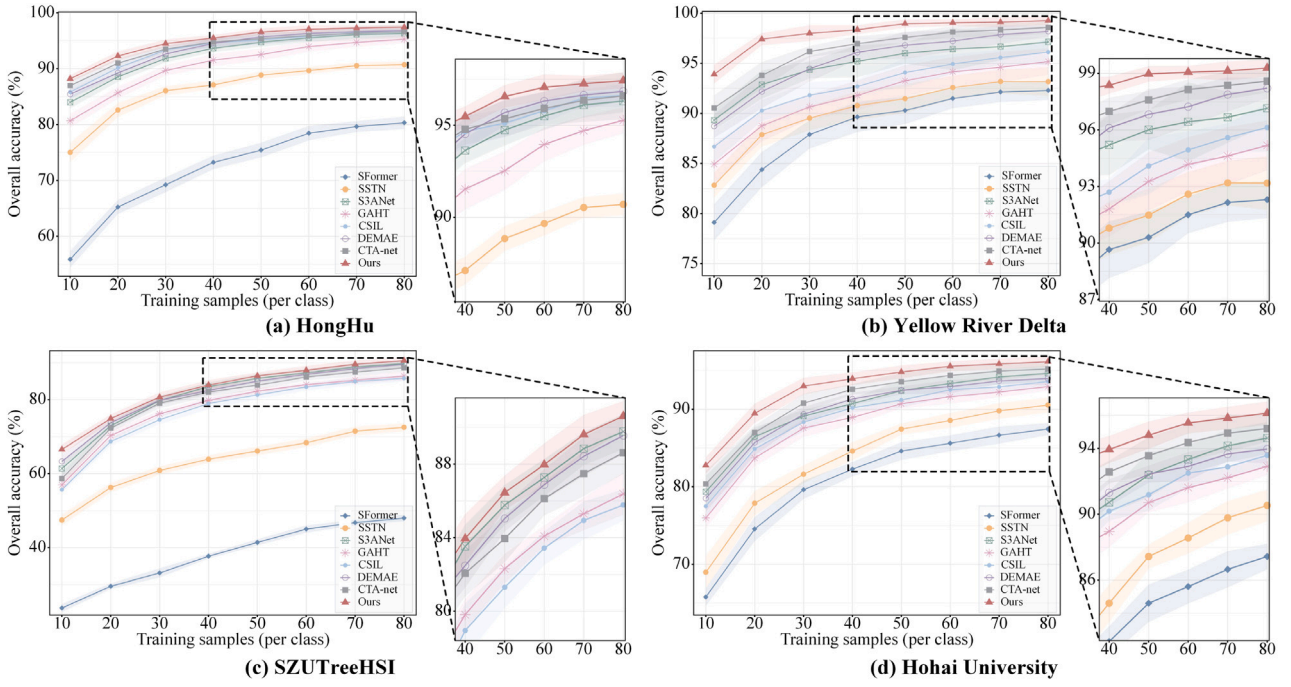


Fig. 14. Performance comparison of different classification methods under various training sample conditions.

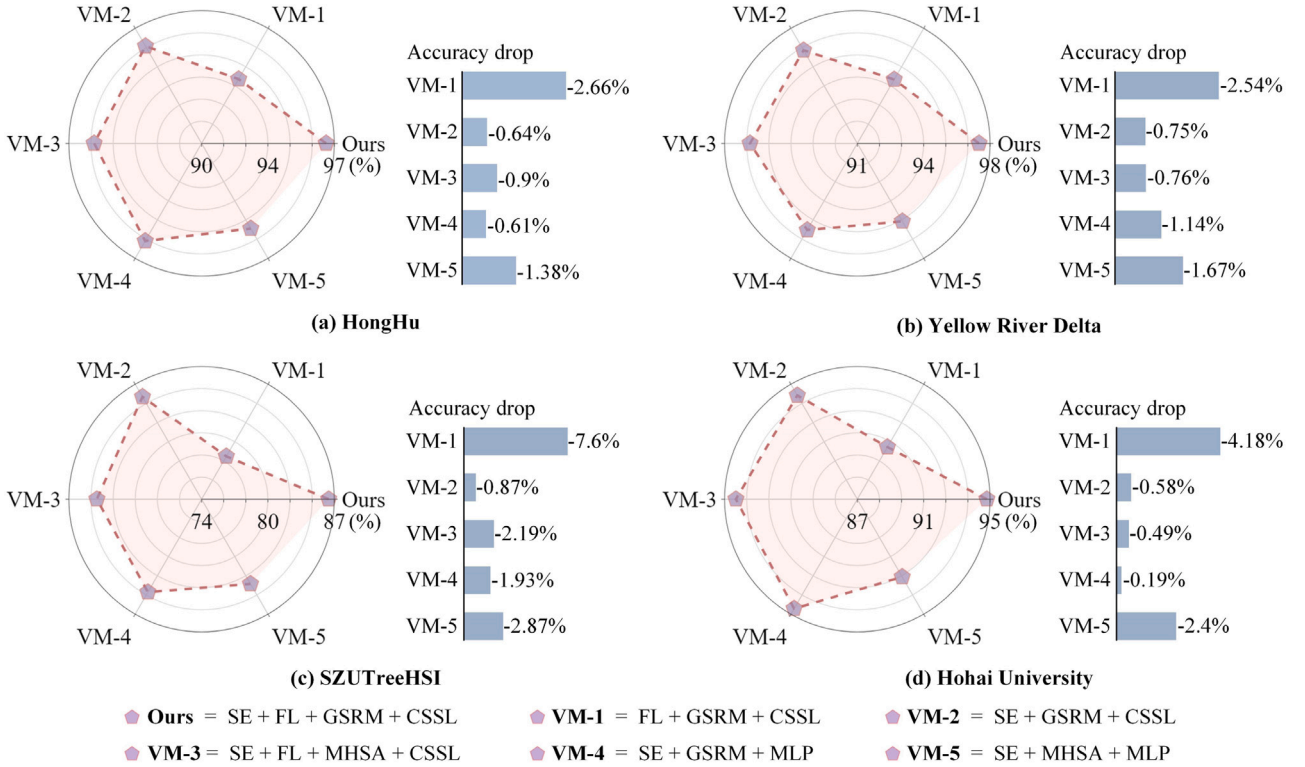


Fig. 15. Comparison of classification performance across different SRTF variant models. VM-1 represents without spectral embedding (SE), VM-2 represents without fidelity loss (FL), VM-3 replaces GSRM with MHSA, VM-4 replaces CSSL with MLP, VM-5 replaces both GSRM and CSSL with MHSA and MLP.

6.2.2. Inference mechanism of SRM

The effectiveness of the proposed SRM in improving classification performance over SAM has been thoroughly demonstrated in the ablation study. In this section, we further compare and analyze how SRM

and SAM capture dependencies among different tokens by examining their respective weight matrices. Fig. 16 presents the normalized weight matrices of SRM and SAM, visualized as heatmaps. Each column in the heatmap represents the association between the token at a specific spatial location and all other tokens, where darker colors

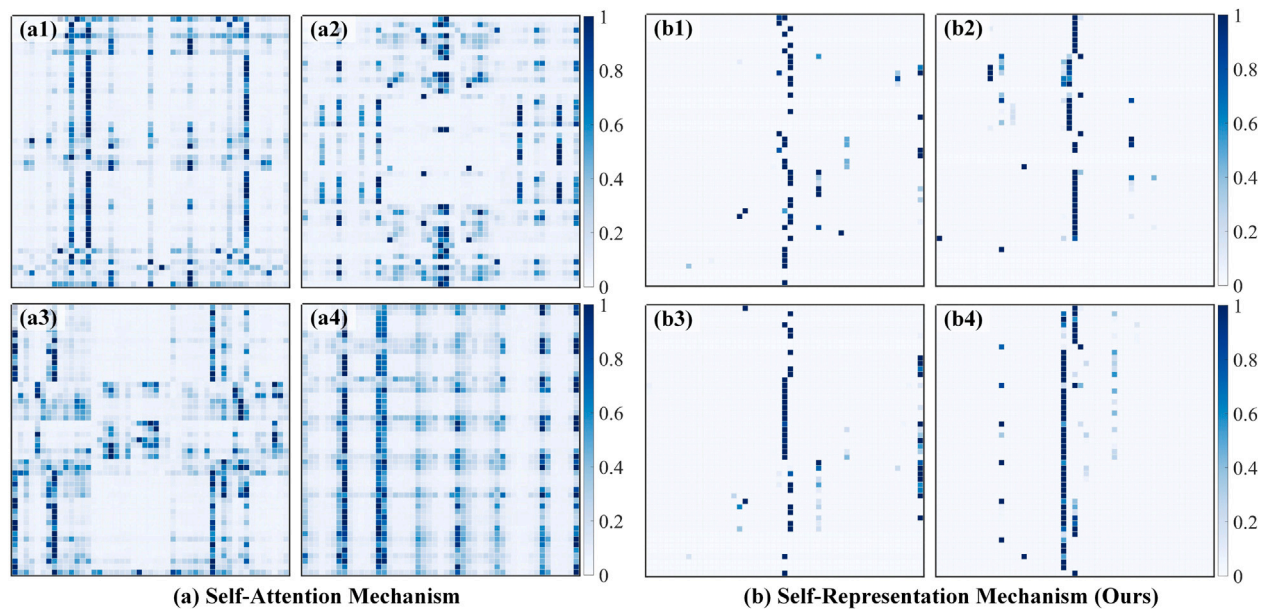


Fig. 16. Comparison of the patterns in modeling token-to-token dependencies between the self-attention mechanism (SAM) and the proposed self-representation mechanism (SRM), each column represents the association between the token at a specific spatial location and all other tokens. Subplots (a1)-(a4) and (b1)-(b4) correspond to the results for Honghu, Yellow River Delta, SZUTreeHSI, and Hohai University, respectively.

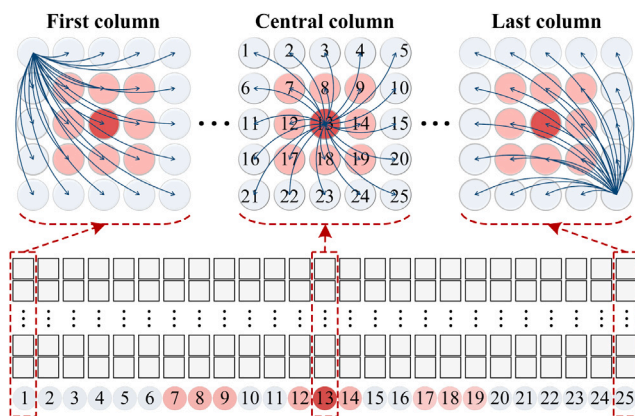


Fig. 17. Visualization of the modeling relationships between tokens represented by each column in the weight matrix.

indicate stronger associations. The specific spatial positions and their corresponding modeled relationships are illustrated in Fig. 17.

From these visual results, two major advantages of SRM can be observed. First, SRM exhibits a distinctively sparse relationship modeling pattern. Sparsity is crucial for effective dependency modeling, as meaningful relationship learning should emphasize informative connections rather than distribute modeling uniformly. Guided by sparse representation principles, the developed SRM naturally performs selective modeling and avoids redundant or noisy dependencies. Second, the dependency patterns produced by SRM better align with the intrinsic requirements of HRSI classification. The classification of HRSI primarily focuses on determining the class of the central pixel within a patch. Accordingly, the key modeling requirement is to capture the dependencies between the central token and surrounding tokens. The heatmaps in Fig. 16(b) show that SRM constructs relationship patterns centered on the central token, effectively capturing the most critical contextual information. In contrast, as shown in Fig. 16(a), SAM generates more diffuse and less focused attention patterns. In summary, by selectively modeling essential dependencies while suppressing redundant relationships, SRM can effectively capture discriminative relationship patterns

within the patch. This targeted and efficient modeling of long-range dependencies ultimately results in more accurate and reliable mapping performance.

6.2.3. Separability of CSSL subspace features

In the proposed SRTF, the subspace features output by the last CSSL module serve directly as the basis for class decisions, making their separability crucial to the final mapping accuracy. To intuitively illustrate the separability of these subspace features, the test samples are fed into the trained model to extract the subspace components from the last CSSL module. The mean vectors of these components are then computed, and dimensionality reduction is performed through the t-SNE algorithm. As shown in Fig. 18, it can be clearly observed that for the four scenes considered in this study, the subspace features of different classes form relatively distinct and well-defined clusters in the low-dimensional space, demonstrating clear separability. Meanwhile, a few samples still exhibit some degree of overlap, which may result from high spectral similarity or the complexity of their spatial distributions, making them harder to distinguish in the feature space. Overall, the subspace features enable effective discrimination between classes, providing a reliable basis for subsequent class decision-making.

6.2.4. Convergence of the loss function

In this section, we analyze the variation trends of the label loss and fidelity loss with the increase of training epochs to verify the convergence of SRTF. Under the setting of 200 training epochs and a learning rate of $1e-4$, Fig. 19 illustrates the convergence of these two loss functions across four different datasets. In terms of the detailed trends, both the label loss and fidelity loss exhibit a rapid decline across all datasets during the first 50 training epochs, followed by a gradually slower decrease between epochs 50 and 150, and finally stabilize in the last 50 epochs. This process not only demonstrates that the model achieves stable and effective optimization across different datasets, but also confirms the rationality of the designed loss functions and their parameter settings.

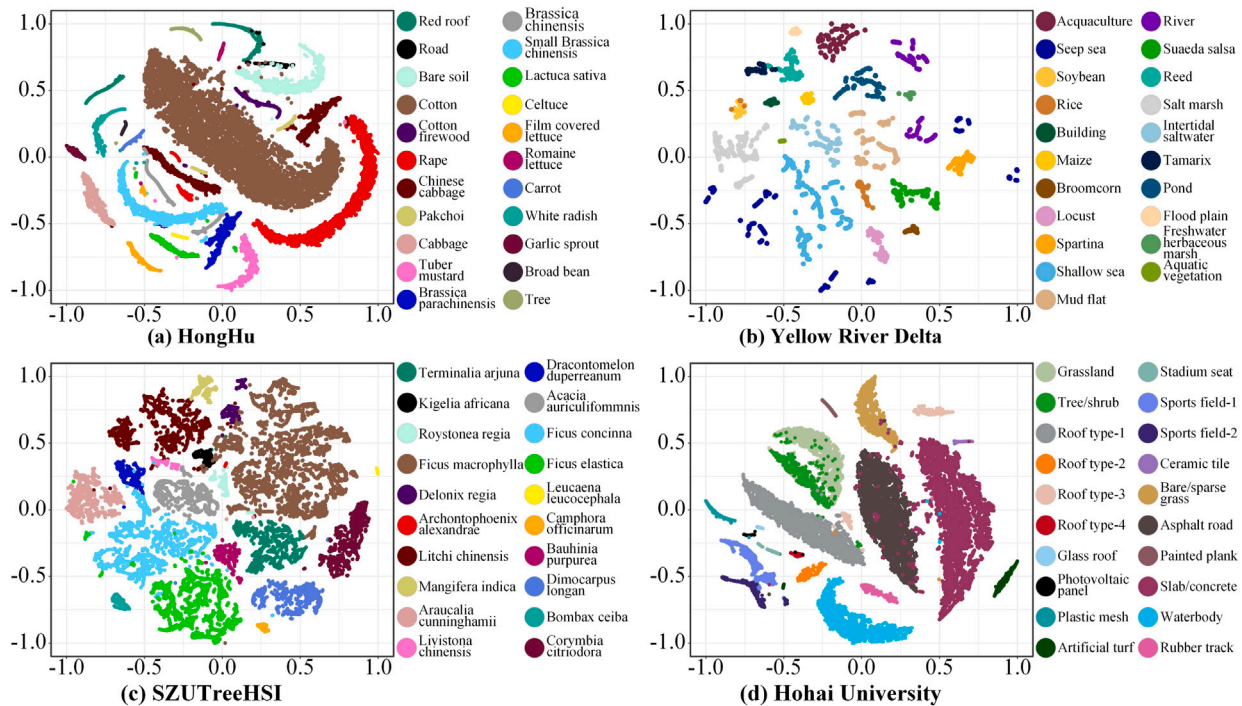


Fig. 18. Visualization of the separability of subspace features from test samples extracted by the last CSSL module.

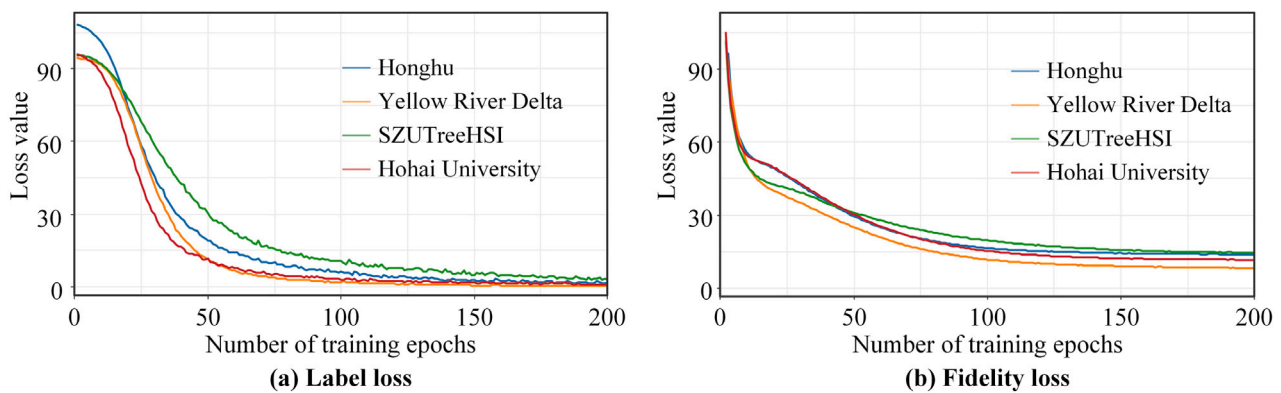


Fig. 19. Convergence of (a) label loss and (b) fidelity loss over training epochs.

6.3. Remaining limitations

Although the proposed SRTF demonstrates excellent performance in multi-scenario land cover mapping while providing interpretability, certain limitations still remain. On the one hand, the model is not fully lightweight and contains a large number of parameters. This can result in high computational and memory costs, especially for large-scale HRSI mapping tasks or under resource-limited conditions. On the other hand, SRTF has been validated only on single-modality HRSI datasets. Its robustness and generalization to multi-modal or multi-source datasets, such as combinations of hyperspectral, LiDAR, or SAR data, are still unknown. Therefore, future work will focus on lightweight optimization, extension to multi-modal and multi-source data, and further enhancement of adaptability and robustness to complex environments and large-scale datasets.

7. Conclusion

This study aims to address the limited interpretability of current Transformer-based methods for HRSI land cover mapping, as they

predominantly depend on empirical architectural designs. By explicitly integrating the theoretical principles of representation models into the Transformer architecture, a novel SRTF method is presented. This design enhances interpretability while maintaining strong practical performance. SRTF incorporates a spectral embedding module to compress redundant information and unify inputs of different patch sizes. Guided by the optimization principle of the sparse representation and subspace decomposition process, the SRM and CSSL modules effectively replace the empirically designed SAM and MLP. Moreover, the decision-making process is aligned with the classification mechanism of representation models, eliminating the need for a semantic-level class token and improving traceability. Experiment results across diverse and challenging study areas demonstrate that SRTF significantly outperforms related methods in both mapping accuracy and visualization quality. A series of detailed analyses further validates the effectiveness and superiority of the proposed method. For future work, we plan to address the remaining limitations by exploring lightweight model designs and multi-modal data fusion, improving computational efficiency and extending the applicability of SRTF to diverse data modalities.

CRediT authorship contribution statement

Xiangyu Nie: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Zhaohui Xue:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization. **Jiahang Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Belgiu, M., Drăguț, L., 2016. Random forest in remote sensing: A review of applications and future directions. *ISPRS J. Photogramm. Remote Sens.* 114, 24–31.
- Bioucas-Dias, J.M., Plaza, A., Camps-Valls, G., Scheunders, P., Nasrabadi, N., Chanussot, J., 2013. Hyperspectral remote sensing data analysis and future challenges. *IEEE Geosci. Remote. Sens. Mag.* 1 (2), 6–36.
- Brown, C.F., Brumby, S.P., Guzder-Williams, B., Birch, T., Hyde, S.B., Mazzariello, J., Czerwinski, W., Pasquarella, V.J., Haertel, R., Ilyushchenko, S., Schwehr, K., Weiss, M., Stolle, F., Hanson, C., Guinan, O., Moore, R., Tait, A.M., 2022. Dynamic World, Near real-time global 10 m land use land cover mapping. *Sci. Data* 9 (1), 251.
- Cheng, S., Chan, R., Du, A., 2024. CACFTNet: A hybrid cov-attention and cross-layer fusion transformer network for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* 62, 1–17.
- Dong, Y., Liu, Q., Du, B., Zhang, L., 2022. Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification. *IEEE Trans. Image Process.* 31, 1559–1572.
- Du, P., Xia, J., Zhang, W., Tan, K., Liu, Y., Liu, S., 2012. Multiple classifier system for remote sensing image classification: A review. *Sensors* 12 (4), 4764–4792.
- Friedl, M.A., McIver, D.K., Hodges, J.C.F., Zhang, X.Y., Muchoney, D., Strahler, A.H., Woodcock, C.E., Gopal, S., Schneider, A., Cooper, A., Baccini, A., Gao, F., Schaaf, C., 2002. Global land cover mapping from MODIS: Algorithms and early results. *Remote Sens. Environ.* 83 (1), 287–302.
- Fu, C., Zhou, T., Guo, T., Zhu, Q., Luo, F., Du, B., 2025. CNN-transformer and channel-spatial attention based network for hyperspectral image classification with few samples. *Neural Netw.* 186, 107283.
- Gong, P., Liu, H., Zhang, M., Li, C., Wang, J., Huang, H., Clinton, N., Ji, L., Li, W., Bai, Y., Chen, B., Xu, B., Zhu, Z., Yuan, C., Ping Suen, H., Guo, J., Xu, N., Li, W., Zhao, Y., Yang, J., Yu, C., Wang, X., Fu, H., Yu, L., Dronova, I., Hui, F., Cheng, X., Shi, X., Xiao, F., Liu, Q., Song, L., 2019. Stable classification with limited sample: Transferring a 30-m resolution sample set collected in 2015 to mapping 10-m resolution global land cover in 2017. *Sci. Bull.* 64 (6), 370–373.
- Gregor, K., View Profile, LeCun, Y., View Profile, 2010. Learning fast approximations of sparse coding. In: *Proceedings of the 27th International Conference on International Conference on Machine Learning*. In: *Guide Proceedings*, pp. 399–406.
- Han, B., Jia, Y., Liu, H., Hou, J., 2025. Irregular tensor low-rank representation for hyperspectral image representation. *IEEE Trans. Image Process.* 34, 3239–3252.
- He, Q., Sun, X., Yan, Z., Wang, B., Zhu, Z., Diaoy, W., Yang, M.Y., 2023. AST: Adaptive self-supervised transformer for optical remote sensing representation. *ISPRS J. Photogramm. Remote Sens.* 200, 41–54.
- He, Z., Wang, P., Zou, Y., Huang, B., Zhu, D., Lee, H., Leung, H., 2026. Dadigan: A dual attention blocks-based disentangled iterative generative adversarial network for cloud and shadow removal on sar and optical images. *Inform. Fusion* 125, 103487. <http://dx.doi.org/10.1016/j.inffus.2025>.
- Hong, D., Han, Z., Yao, J., Gao, L., Zhang, B., Plaza, A., Chanussot, J., 2022. SpectralFormer: Rethinking hyperspectral image classification with transformers. *IEEE Trans. Geosci. Remote Sens.* 60, 1–15.
- Hu, C., Jia, S., Zhang, F., Xiao, C., Ruan, M., Thrasher, J., Li, X., 2023. UPDEXplainer: An interpretable transformer-based framework for urban physical disorder detection using street view imagery. *ISPRS J. Photogramm. Remote Sens.* 204, 209–222.
- Hu, X., Wang, X., Zhong, Y., Zhang, L., 2022. S3ANet: Spectral-spatial-scale attention network for end-to-end precise crop classification based on UAV-borne H2 imagery. *ISPRS J. Photogramm. Remote Sens.* 183, 147–163.
- Huang, H., Duan, Y., He, H., Shi, G., Luo, F., 2019. Spatial-spectral local discriminant projection for dimensionality reduction of hyperspectral image. *ISPRS J. Photogramm. Remote Sens.* 156, 77–93.
- Huang, Y., Peng, J., Chen, N., Sun, W., Du, Q., Ren, K., Huang, K., 2023a. Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network. *ISPRS J. Photogramm. Remote Sens.* 203, 37–54.
- Huang, Y., Peng, J., Chen, N., Sun, W., Du, Q., Ren, K., Huang, K., 2023b. Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network. *ISPRS J. Photogramm. Remote Sens.* 203, 37–54.
- Kuang, X., Xiang, S., Guo, J., 2025. Soil moisture retrieval and trend prediction using multi-temporal remote sensing data: An interpretable deep regression approach. *Expert Syst. Appl.* 287, 128172.
- Kumar, B., Dikshit, O., Gupta, A., Singh, M.K., 2020. Feature extraction for hyperspectral image classification: A review. *Int. J. Remote Sens.* 41 (16), 6248–6287.
- Kumar, V., Singh, R.S., Rambabu, M., Dua, Y., 2024. Deep learning for hyperspectral image classification: A survey. *Comput. Sci. Rev.* 53, 100658.
- Li, N., Jiang, S., Xue, J., Ye, S., Jia, S., 2024. Texture-Aware self-attention model for hyperspectral tree species classification. *IEEE Trans. Geosci. Remote Sens.* 62, 1–15.
- Li, W., Liu, J., Du, Q., 2016. Sparse and Low-rank graph for discriminant analysis of hyperspectral imagery. *IEEE Trans. Geosci. Remote Sens.* 54 (7), 4094–4105.
- Li, X., Wen, C., Hu, Y., Yuan, Z., Zhu, X.X., 2024. Vision-language models in remote sensing: Current progress and future trends. *IEEE Geosci. Remote. Sens. Mag.* 12 (2), 32–66.
- Li, Z., Xue, Z., Jia, M., Nie, X., Wu, H., Zhang, M., Su, H., 2024. DEMAEE: Diffusion-enhanced masked autoencoder for hyperspectral image classification with few labeled samples. *IEEE Trans. Geosci. Remote Sens.* 62, 1–16.
- Liu, R., Ling, J., Zhang, H., 2024. SoftFormer: SAR-optical fusion transformer for urban land use and land cover classification. *ISPRS J. Photogramm. Remote Sens.* 218, 277–293.
- Lou, C., Al-qaness, M.A.A., Al-Alimi, D., Dahou, A., Abd Elaziz, M., Abualigah, L., Ewees, A.A., 2025. Land Use/Land Cover (LULC) classification using hyperspectral images: A review. *Geo-Spatial Inf. Sci.* 28 (2), 345–386.
- Mei, S., Song, C., Ma, M., Xu, F., 2022. Hyperspectral image classification using group-aware hierarchical transformer. *IEEE Trans. Geosci. Remote Sens.* 60, 1–14.
- Moharram, M.A., Sundaram, D.M., 2023. Land use and land cover classification with hyperspectral data: A comprehensive review of methods, challenges and future directions. *Neurocomputing* 536, 90–113.
- Patro, R.N., Subudhi, S., Biswal, P.K., Dell'acqua, F., 2021. A review of unsupervised band selection techniques: Land cover classification for hyperspectral earth observation data. *IEEE Geosci. Remote. Sens. Mag.* 9 (3), 72–111.
- Peng, J., Du, Q., 2017. Robust joint sparse representation based on maximum correntropy criterion for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* 55 (12), 7152–7164.
- Peng, J., Sun, W., Li, H.-C., Li, W., Meng, X., Ge, C., Du, Q., 2022. Low-rank and sparse representation for hyperspectral image processing: A review. *IEEE Geosci. Remote. Sens. Mag.* 10 (1), 10–43.
- Shi, C., Chen, P., Fang, L., Zhao, M., Hei, X., Miao, Q., 2025. Multiscale spatial-frequency-domain cross-transformer for hyperspectral image classification. *IEEE Trans. Instrum. Meas.* 74, 1–15.
- Song, X.-P., Hansen, M.C., Stehman, S.V., Potapov, P.V., Tyukavina, A., Vermote, E.F., Townshend, J.R., 2018. Global land change 1982–2016. *Nature* 560 (7720), 639–643.
- Song, Y., Zhang, J., Liu, Z., Xu, Y., Quan, S., Sun, L., Bi, J., Wang, X., 2025. Deep learning for hyperspectral image classification: A comprehensive review and future predictions. *Inf. Fusion* 123, 103285.
- Su, H., Shao, F., Gao, Y., Zhang, H., Sun, W., Du, Q., 2023. Probabilistic collaborative representation based ensemble learning for classification of wetland hyperspectral imagery. *IEEE Trans. Geosci. Remote Sens.* 61, 1–17.
- Su, H., Yao, W., Wu, Z., Zheng, P., Du, Q., 2021. Kernel low-rank representation with elastic net for China coastal wetland land cover classification using GF-5 hyperspectral imagery. *ISPRS J. Photogramm. Remote Sens.* 171, 238–252.
- Vali, A., Comai, S., Matteucci, M., 2020. Deep learning for land use and land cover classification based on hyperspectral and multispectral earth observation data: A review. *Remote Sens.* 12 (15), 2495.
- Verburg, P.H., Neumann, K., Nol, L., 2011. Challenges in using land use and land cover data for global change studies. *Global Change Biol.* 17 (2), 974–989.
- Wang, P., Liu, Y., Liang, X., Zhu, D., Gong, X., Ye, Y., Lee, H., Huang, B., 2025a. Cirsnet: A cyclic registration network for sar and optical images. *IEEE Trans. Geosci. Remote Sens.* 63, 1–19. <http://dx.doi.org/10.1109/TGRS.2025.3540258>.
- Wang, Y., Sun, Y., Cao, X., Wang, Y., Zhang, W., Cheng, X., 2023. A review of regional and Global scale Land Use/Land Cover (LULC) mapping products generated from satellite remote sensing. *ISPRS J. Photogramm. Remote Sens.* 206, 311–334.
- Wang, W., Sun, Y., Li, J., Wang, X., 2024. Frequency and spatial based multi-layer context network (FSCNet) for remote sensing scene classification. *Int. J. Appl. Earth Obs. Geoinf.* 128, 103781.

- Wang, S., Yuan, G., Li, J., 2025. RTDU: Interpretable region-aware transformer-based deep unfolding network for pan-sharpening. *IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens.* 18, 12897–12914.
- Winkler, K., Fuchs, R., Rounsevell, M., Herold, M., 2021. Global land use changes are four times greater than previously estimated. *Nat. Commun.* 12 (1), 2501.
- Wittstruck, L., Waske, B., Jarmer, T., 2025. Multi-modal vision transformer for high-resolution soil texture prediction of German agricultural soils using remote sensing imagery. *Remote Sens. Environ.* 331, 114985.
- Xia, J., Chanussot, J., Du, P., He, X., 2016. Rotation-based support vector machine ensemble in classification of hyperspectral data with limited training samples. *IEEE Trans. Geosci. Remote Sens.* 54 (3), 1519–1531.
- Xia, S., Zhang, X., Meng, H., Jiao, L., 2025. A novel transformer-KAN network utilizes multimodality contrastive learning and mask reconstruction for remote sensing classification. *IEEE Trans. Geosci. Remote Sens.* 63, 1–16.
- Xue, Z., Du, P., Li, J., Su, H., 2017. Sparse graph regularization for robust crop mapping using hyperspectral remotely sensed imagery with very few in situ data. *ISPRS J. Photogramm. Remote Sens.* 124, 1–15.
- Yang, J., Du, B., Zhang, L., 2023. From center to surrounding: An interactive learning framework for hyperspectral image classification. *ISPRS J. Photogramm. Remote Sens.* 197, 145–166.
- Yu, Y., Buchanan, S., Pai, D., Chu, T., Wu, Z., Tong, S., Bai, H., Zhai, Y., Haeffele, B.D., Ma, Y., 2024. White-box transformers via sparse rate reduction: Compression is all there is? *J. Mach. Learn. Res.* 25 (300), 1–128.
- Zhang, P., Lu, B., Shang, J., Tan, C., Xu, Q., Shi, L., Jin, S., Wang, X., Jiang, Y., Yang, Y., Zang, H., Ge, J., Zeng, Z., 2025. TKSF-KAN: Transformer-enhanced oat yield modeling and transferability across major oat-producing regions in China using UAV multisource data. *ISPRS J. Photogramm. Remote Sens.* 224, 166–186.
- Zhang, Y., Ma, Y., Dai, X., Li, H., Mei, X., Ma, J., 2021. Locality-constrained sparse representation for hyperspectral image classification. *Inform. Sci.* 546, 858–870.
- Zhang, Y., Xue, X., Wang, T., He, M., 2017. A hybrid subpixel mapping framework for hyperspectral images using collaborative representation. *IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens.* 10 (11), 5073–5086.
- Zhao, Z., Xu, X., Li, S., Plaza, A., 2024. Hyperspectral image classification using groupwise separable convolutional vision transformer network. *IEEE Trans. Geosci. Remote Sens.* 62, 1–17.
- Zhong, Y., Hu, X., Luo, C., Wang, X., Zhao, J., Zhang, L., 2020. WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF. *Remote Sens. Environ.* 250, 112012.
- Zhong, Z., Li, Y., Ma, L., Li, J., Zheng, W.-S., 2021. Spectral-spatial transformer network for hyperspectral image classification: A factorized architecture search framework. *IEEE Trans. Geosci. Remote Sens.* 60, 1–15.